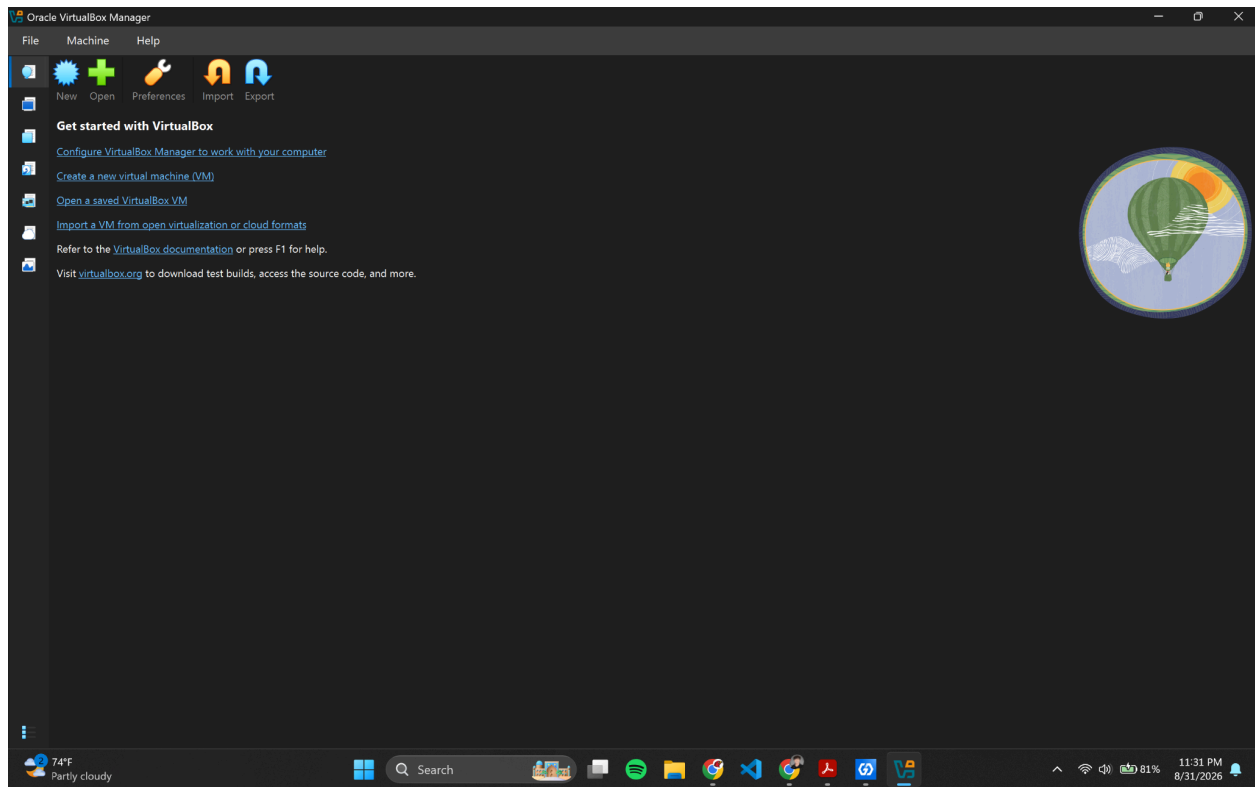
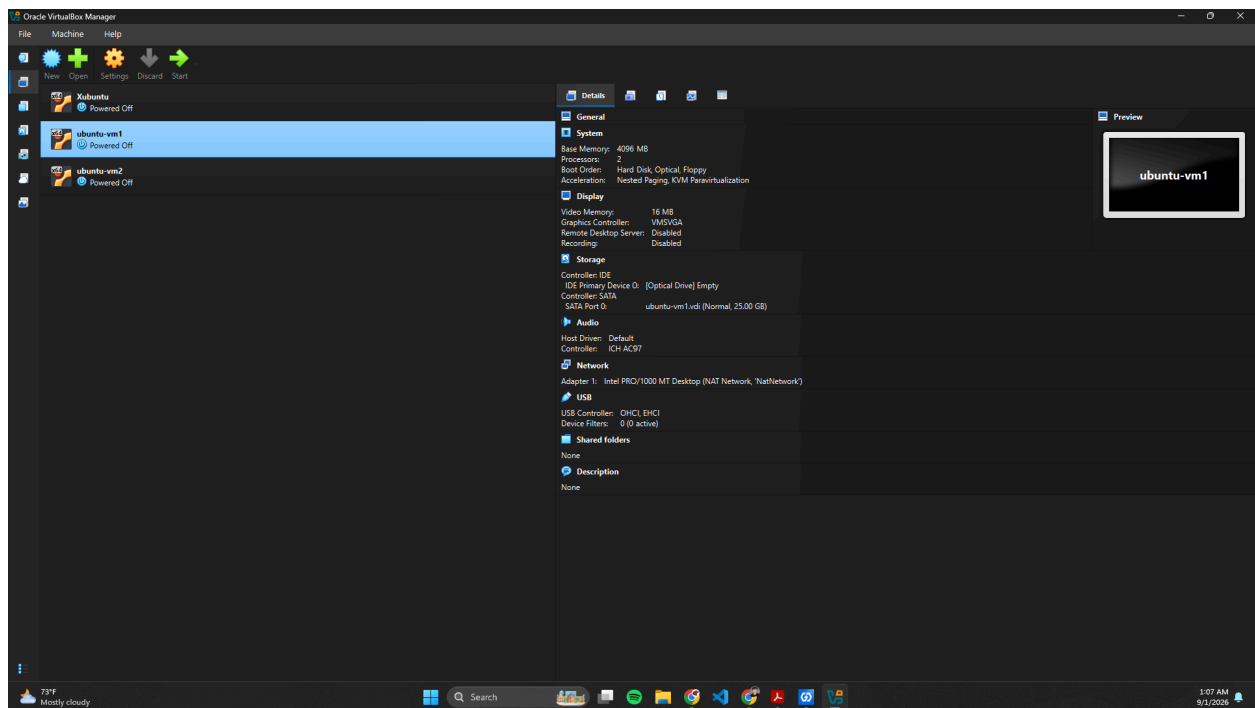
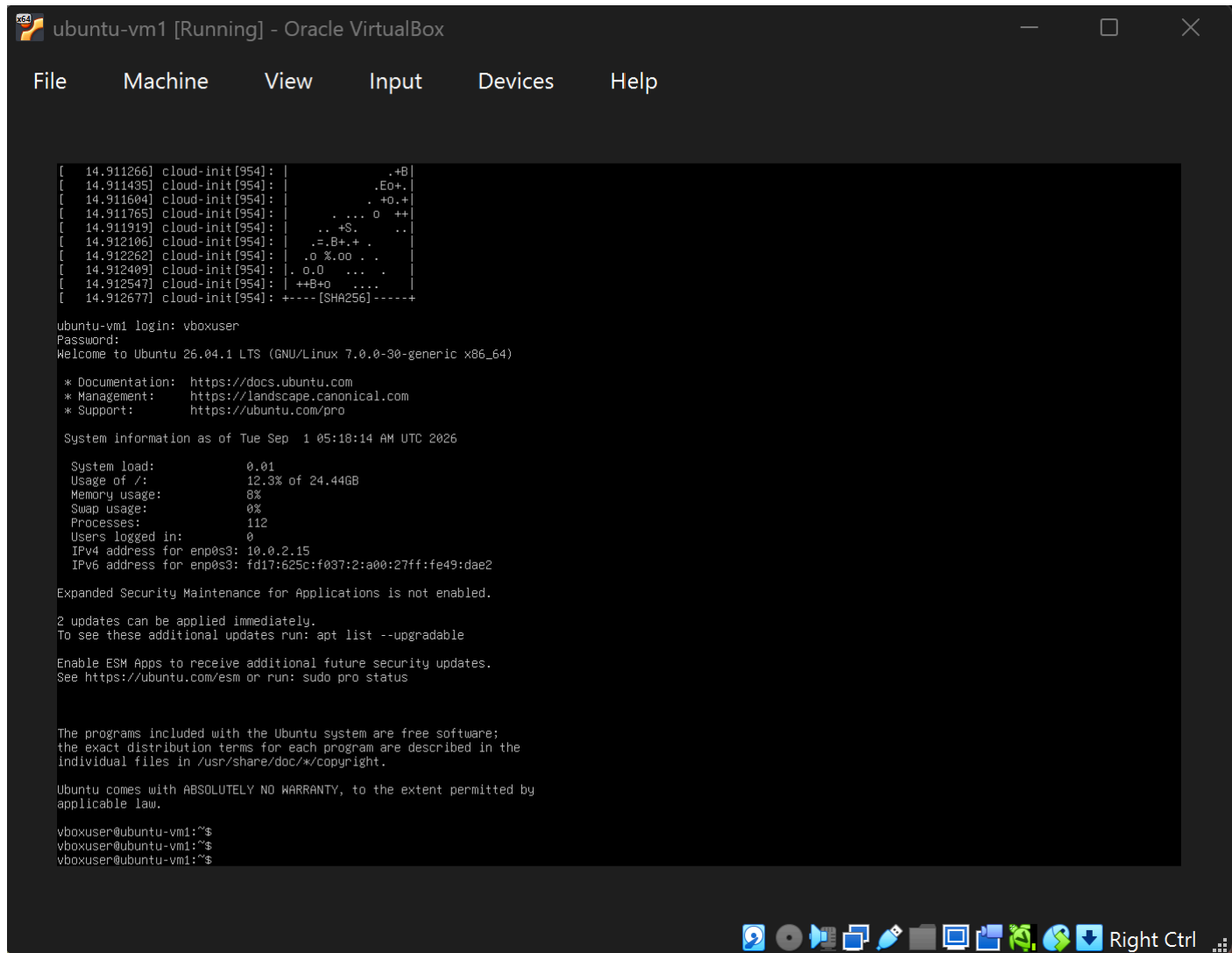


1b



1e





ghi

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

* Management: https://landscape.canonical.com
* Support: https://ubuntu.com/pro

System Information as of Tue Sep 1 05:37:05 AM UTC 2026

System load: 0.08
Usage of /: 12.4% of 24.44GB
Memory usage: 7%
Swap usage: 0%
Processes: 116
Users logged in: 0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe49:dae2

Expanded Security Maintenance for Applications is not enabled.

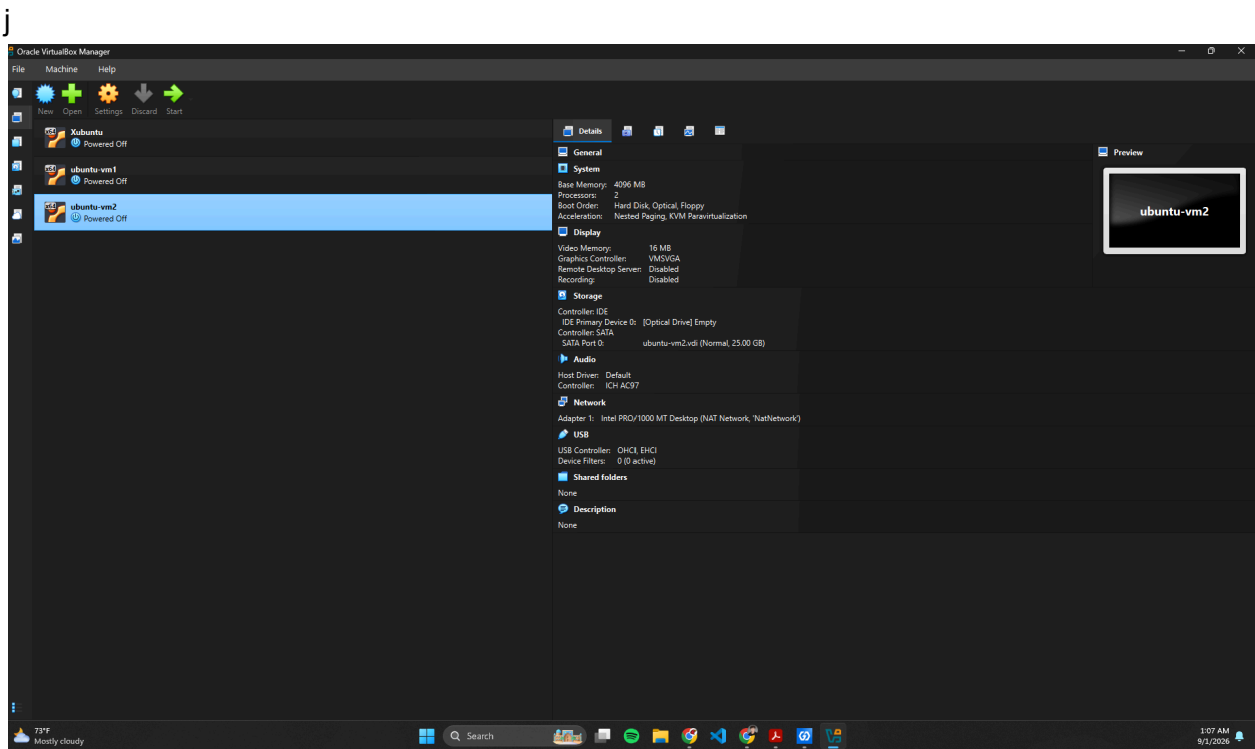
2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

vboxuser@ubuntu-vm1:~$ sudo ufw status
[sudo: authenticate] Password:
Status: inactive
vboxuser@ubuntu-vm1:~$ sudo ufw default deny incoming
Default incoming policy changed to 'deny'
(be sure to update your rules accordingly)
vboxuser@ubuntu-vm1:~$ sudo ufw default allow outgoing
Default outgoing policy changed to 'allow'
(be sure to update your rules accordingly)
vboxuser@ubuntu-vm1:~$ sudo ufw allow ssh
Rules updated
Rules updated (v6)
vboxuser@ubuntu-vm1:~$ sudo ufw enable
Firewall is active and enabled on system startup
vboxuser@ubuntu-vm1:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)

vboxuser@ubuntu-vm1:~$
```



```
vboxuser@ubuntu-vm2:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)

vboxuser@ubuntu-vm2:~$ _
```

1k/1

```
vboxuser@ubuntu-vm2:~$ ssh vboxuser@10.0.2.3
vboxuser@10.0.2.3's password:
Permission denied, please try again.
vboxuser@10.0.2.3's password:
vboxuser@ubuntu-vm2:~$ ssh vboxuser@10.0.2.3
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

+ Documentation:  https://docs.ubuntu.com
+ Management:    https://landscape.canonical.com
+ Support:       https://ubuntu.com/pro

System information as of Tue Sep  1 06:26:14 AM UTC 2026

System load:  0.13          Processes:      124
Usage of /:   12.6% of 24.44GB Users logged in:  0
Memory usage: 10%          IPv4 address for enp0s3: 10.0.2.3
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

vboxuser@ubuntu-vm1:~$ _
```

```
vboxuser@ubuntu-vm1:~$ ssh vboxuser@10.0.2.4
vboxuser@10.0.2.4's password:
Permission denied, please try again.
vboxuser@10.0.2.4's password:
vboxuser@ubuntu-vm1:~$ ssh vboxuser@10.0.2.4
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

+ Documentation:  https://docs.ubuntu.com
+ Management:    https://landscape.canonical.com
+ Support:       https://ubuntu.com/pro

System information as of Tue Sep  1 06:26:29 AM UTC 2026

System load:  0.04          Processes:      123
Usage of /:   12.6% of 24.44GB Users logged in:  0
Memory usage: 9%           IPv4 address for enp0s3: 10.0.2.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

vboxuser@ubuntu-vm2:~$
```

2a

```
ubuntu-vm1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ ssh vboxuser@10.0.2.4
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-30-generic x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/pro

System information as of Tue Sep  1 02:44:49 PM UTC 2026

System load:  0.11          Processes:      127
Usage of /:   13.2% of 24.44GB Users logged in:  0
Memory usage: 9%           IPv4 address for enp0s3: 10.0.2.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

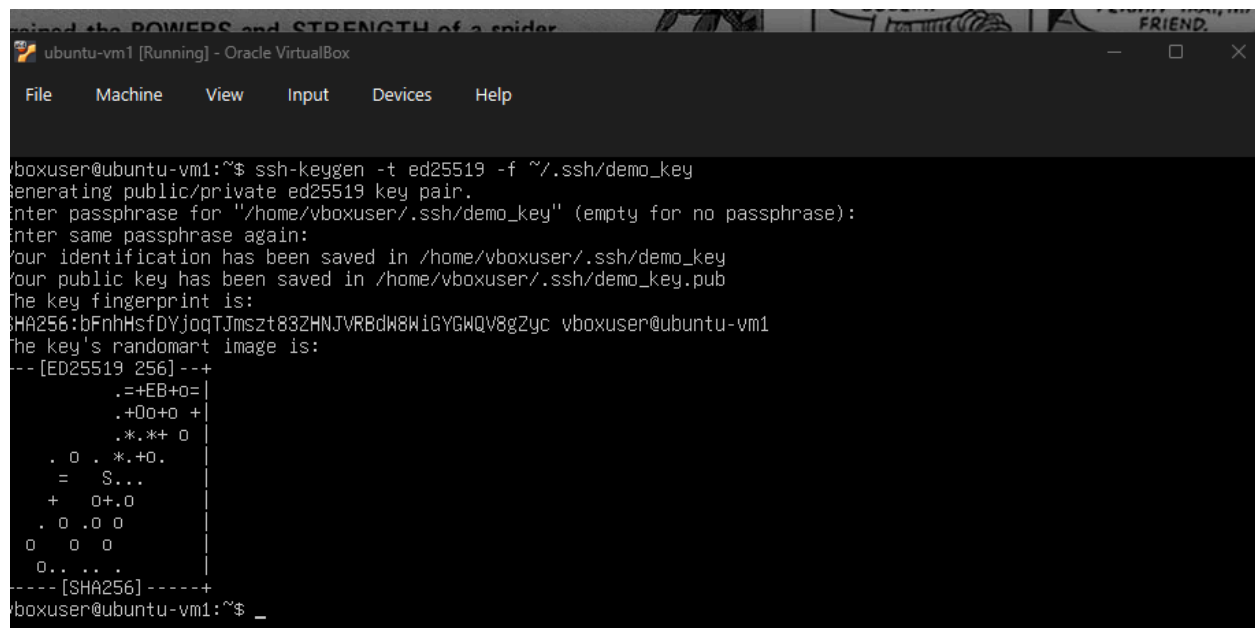
2 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Tue Sep  1 06:26:29 2026 from 10.0.2.3
vboxuser@ubuntu-vm2:~$ _
```

Logs into another machine over encrypted connection

2b

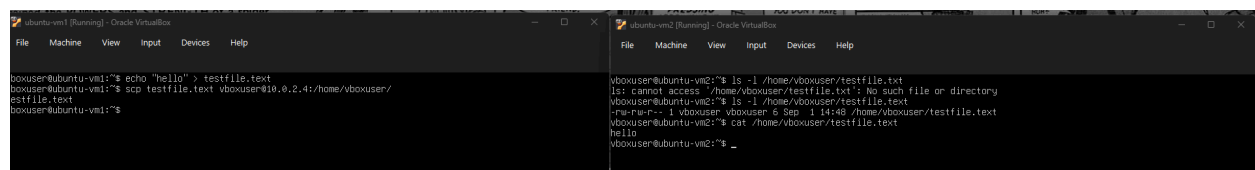


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ ssh-keygen -t ed25519 -f ~/.ssh/demo_key
Generating public/private ed25519 key pair.
Enter passphrase for '/home/vboxuser/.ssh/demo_key' (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vboxuser/.ssh/demo_key
Your public key has been saved in /home/vboxuser/.ssh/demo_key.pub
The key fingerprint is:
SHA256:bFnHsfdYjoqTJmszt83ZHNJVrBdW8WiGYGwQV8gZyc vboxuser@ubuntu-vm1
The key's randomart image is:
-- [ED25519 256] --+
      .+EB+O=|
      .+O+O+ +|
      .*.#+ 0 |
      . 0 . *.+O. |
      = S... |
      + 0+.0 |
      . 0 .0 0 |
      0 0 0 |
      0.. .. |
----- [SHA256] -----+
vboxuser@ubuntu-vm1:~$ _
```

(I generated a demo key because I already had one from before)
Generate private and public key pair for auth.

2c



```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

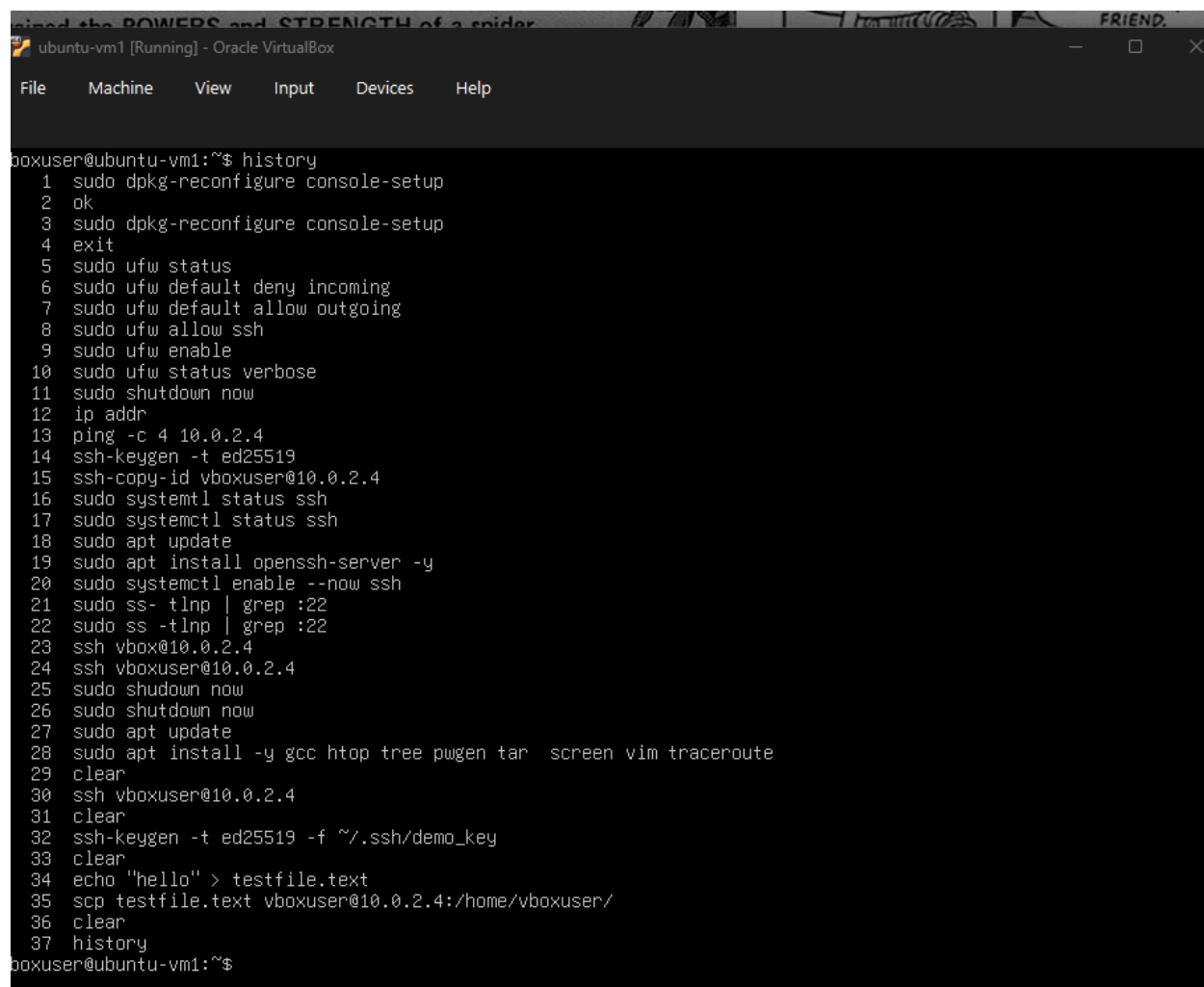
vboxuser@ubuntu-vm1:~$ echo "hello" > testfile.txt
vboxuser@ubuntu-vm1:~$ scp testfile.txt vboxuser@10.0.2.4:/home/vboxuser/
testfile.txt
vboxuser@ubuntu-vm1:~$

ubuntu-vm2 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm2:~$ ls -l /home/vboxuser/testfile.txt
ls: cannot access '/home/vboxuser/testfile.txt': No such file or directory
vboxuser@ubuntu-vm2:~$ ls -l /home/vboxuser/testfile.txt
-rw-r--r-- 1 vboxuser vboxuser 6 Sep 14:48 /home/vboxuser/testfile.txt
vboxuser@ubuntu-vm2:~$ cat /home/vboxuser/testfile.txt
hello
vboxuser@ubuntu-vm2:~$ _
```

Securely copies files between machines over ssh

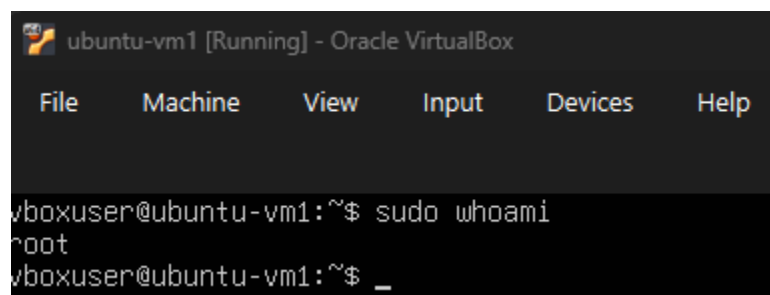
2d



```
boxuser@ubuntu-vm1:~$ history
1  sudo dpkg-reconfigure console-setup
2  ok
3  sudo dpkg-reconfigure console-setup
4  exit
5  sudo ufw status
6  sudo ufw default deny incoming
7  sudo ufw default allow outgoing
8  sudo ufw allow ssh
9  sudo ufw enable
10 sudo ufw status verbose
11 sudo shutdown now
12 ip addr
13 ping -c 4 10.0.2.4
14 ssh-keygen -t ed25519
15 ssh-copy-id vboxuser@10.0.2.4
16 sudo systemctl status ssh
17 sudo systemctl status ssh
18 sudo apt update
19 sudo apt install openssh-server -y
20 sudo systemctl enable --now ssh
21 sudo ss -tlnp | grep :22
22 sudo ss -tlnp | grep :22
23 ssh vbox@10.0.2.4
24 ssh vboxuser@10.0.2.4
25 sudo shutdown now
26 sudo shutdown now
27 sudo apt update
28 sudo apt install -y gcc htop tree pwgen tar screen vim traceroute
29 clear
30 ssh vboxuser@10.0.2.4
31 clear
32 ssh-keygen -t ed25519 -f ~/.ssh/demo_key
33 clear
34 echo "hello" > testfile.text
35 scp testfile.text vboxuser@10.0.2.4:/home/vboxuser/
36 clear
37 history
boxuser@ubuntu-vm1:~$
```

Lists all previously run commands

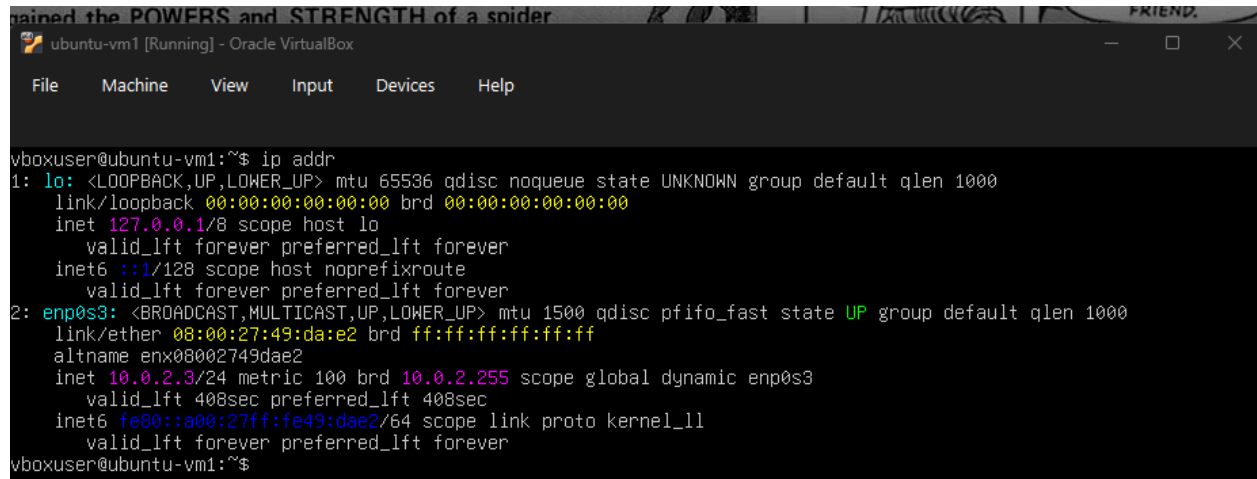
2e



```
vboxuser@ubuntu-vm1:~$ sudo whoami
root
vboxuser@ubuntu-vm1:~$ _
```

Runs command with root privileges

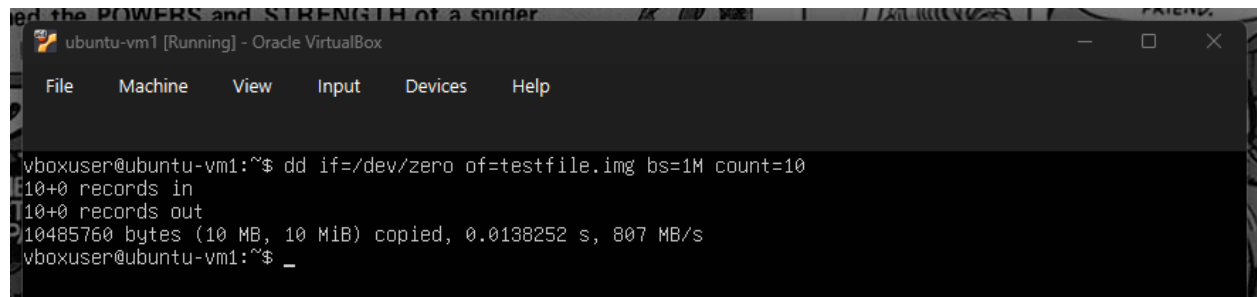
2f



```
vboxuser@ubuntu-vm1:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:49:da:e2 brd ff:ff:ff:ff:ff:ff
    altname enx08002749dae2
    inet 10.0.2.3/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 408sec preferred_lft 408sec
    inet6 fe80::a00:27ff:fe49:dae2/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
vboxuser@ubuntu-vm1:~$
```

Show network interface and addresses

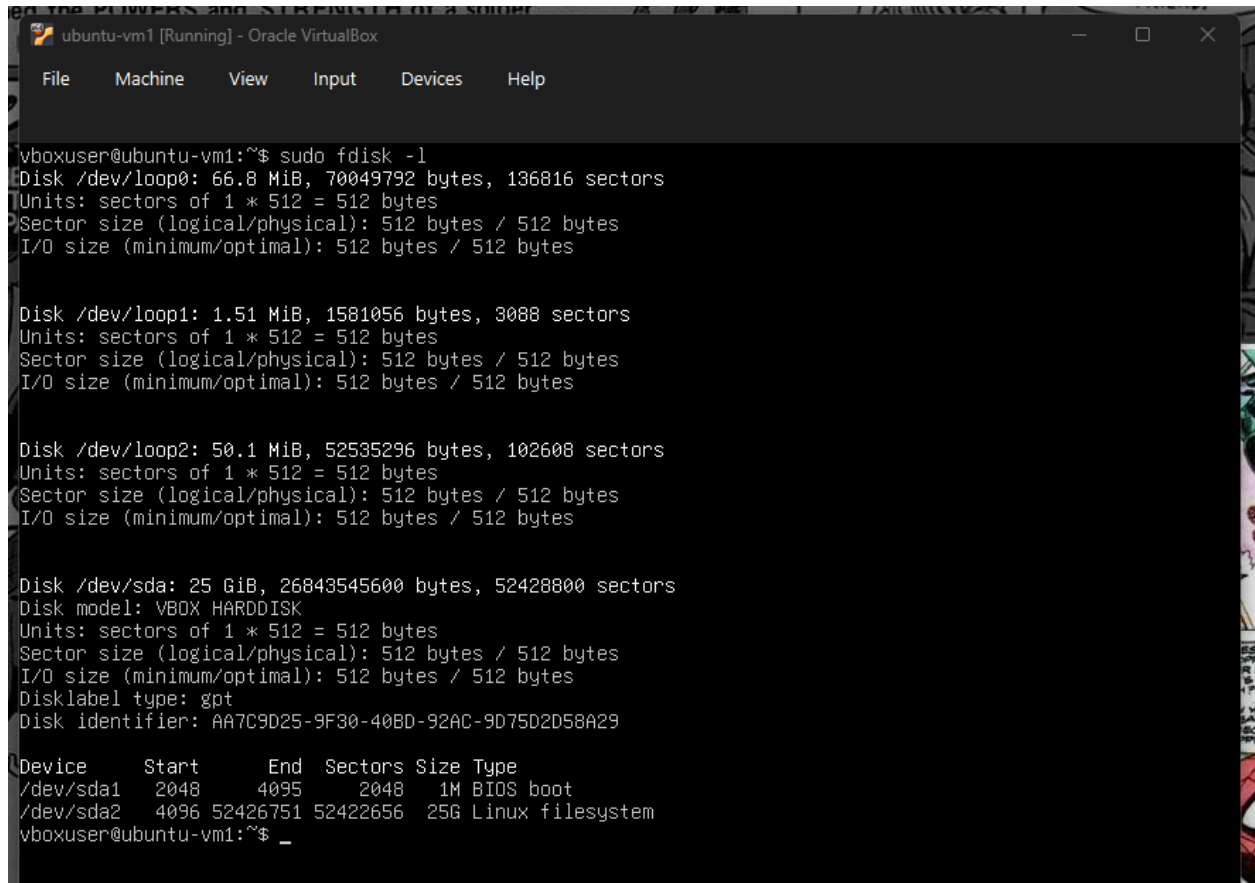
2g



```
vboxuser@ubuntu-vm1:~$ dd if=/dev/zero of=testfile.img bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.0138252 s, 807 MB/s
vboxuser@ubuntu-vm1:~$ _
```

Copiers raw data block by block

2h



```
vboxuser@ubuntu-vm1:~$ sudo fdisk -l
Disk /dev/loop0: 66.8 MiB, 70049792 bytes, 136816 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/loop1: 1.51 MiB, 1581056 bytes, 3088 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

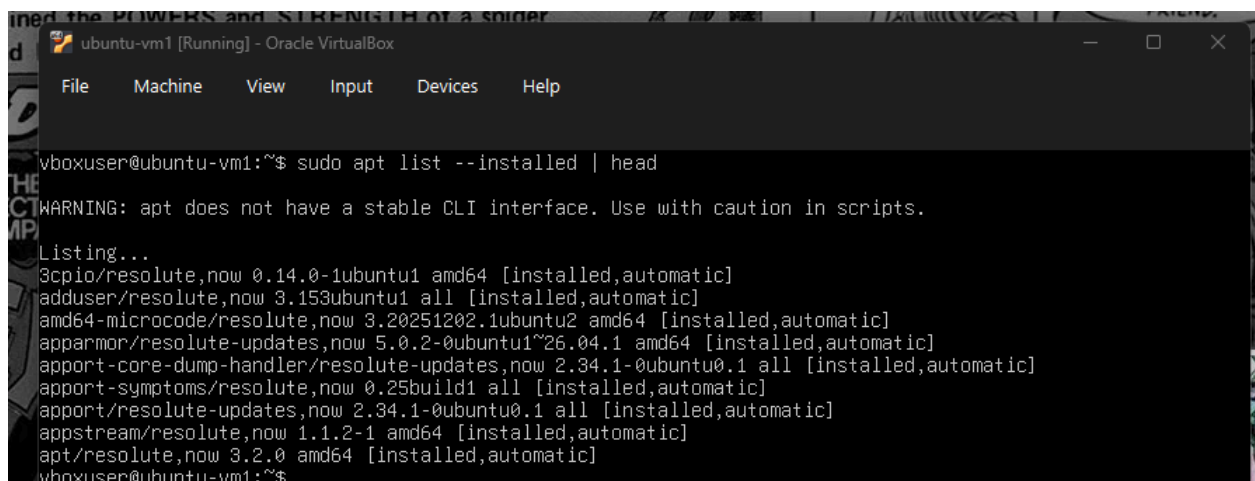
Disk /dev/loop2: 50.1 MiB, 52535296 bytes, 102608 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes

Disk /dev/sda: 25 GiB, 26843545600 bytes, 52428800 sectors
Disk model: VBOX HARDISK
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: gpt
Disk identifier: AA7C9D25-9F30-40BD-92AC-9D75D2D58A29

Device      Start      End  Sectors  Size Type
/dev/sda1    2048      4095     2048    1M BIOS boot
/dev/sda2    4096 52426751 52422656   25G Linux filesystem
vboxuser@ubuntu-vm1:~$ _
```

Shows view of disk partitions table

2i

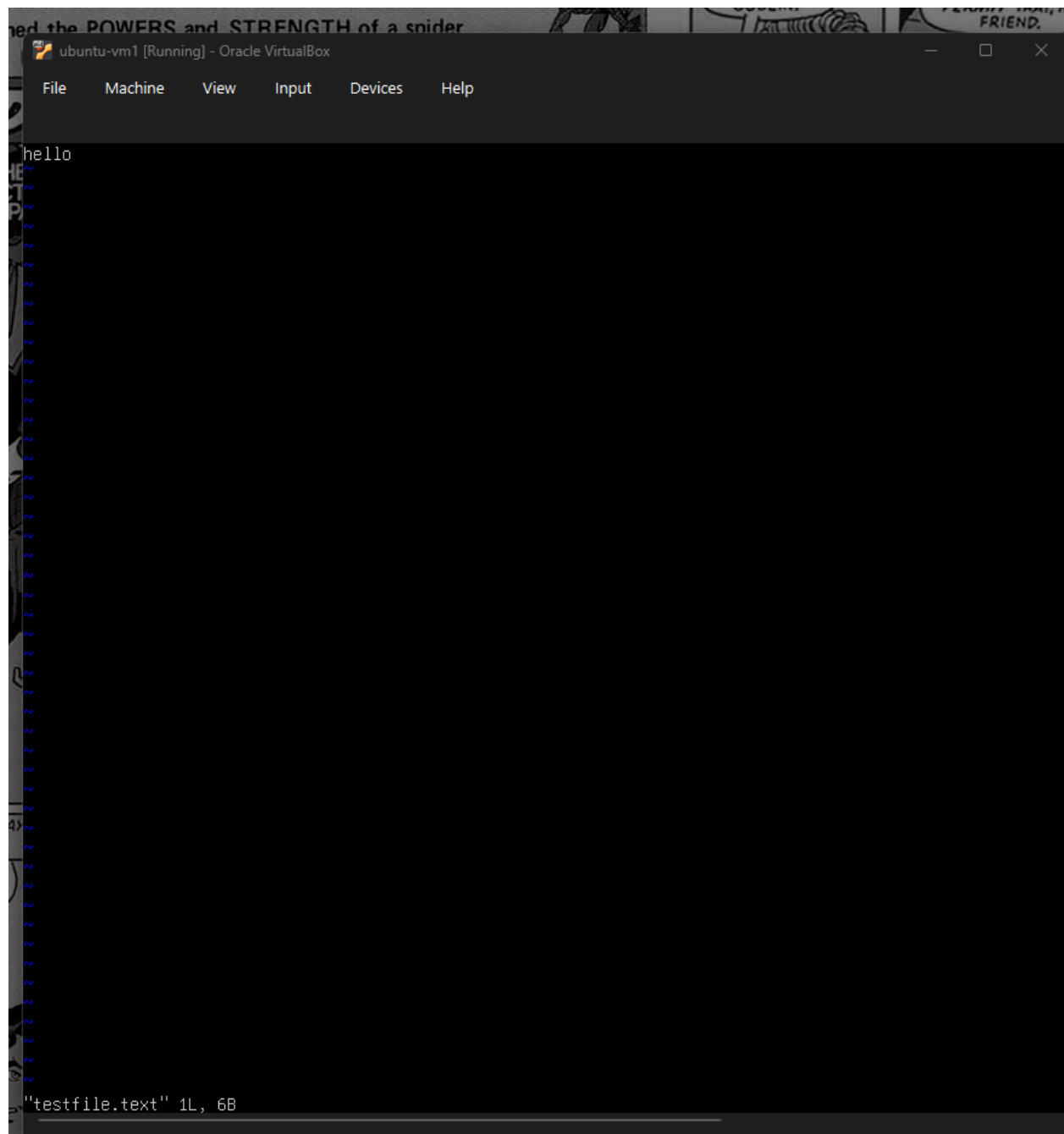


```
vboxuser@ubuntu-vm1:~$ sudo apt list --installed | head
WARNING: apt does not have a stable CLI interface. Use with caution in scripts.

Listing...
3cpio/resolute,now 0.14.0-1ubuntu1 amd64 [installed,automatic]
adduser/resolute,now 3.153ubuntu1 all [installed,automatic]
amd64-microcode/resolute,now 3.20251202.1ubuntu2 amd64 [installed,automatic]
apparmor/resolute-updates,now 5.0.2-0ubuntu1~26.04.1 amd64 [installed,automatic]
appport-core-dump-handler/resolute-updates,now 2.34.1-0ubuntu0.1 all [installed,automatic]
appport-symptoms/resolute,now 0.25build1 all [installed,automatic]
appport/resolute-updates,now 2.34.1-0ubuntu0.1 all [installed,automatic]
appstream/resolute,now 1.1.2-1 amd64 [installed,automatic]
apt/resolute,now 3.2.0 amd64 [installed,automatic]
vboxuser@ubuntu-vm1:~$
```

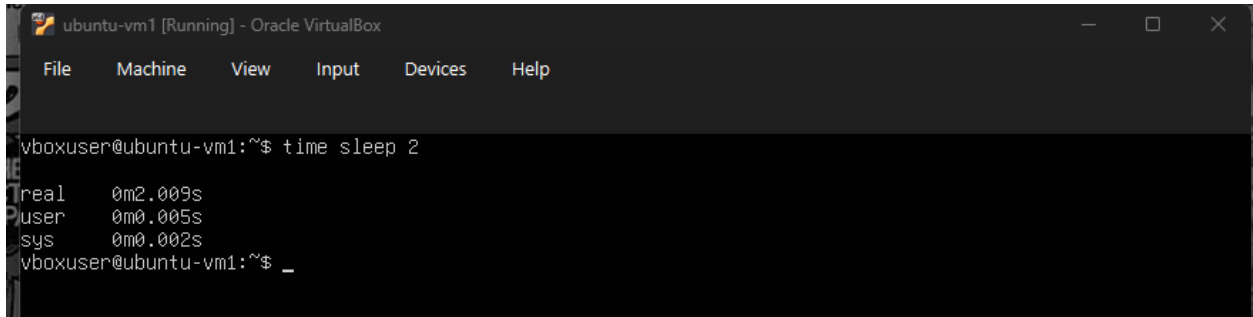
Package manager

2j



Text editor

2k

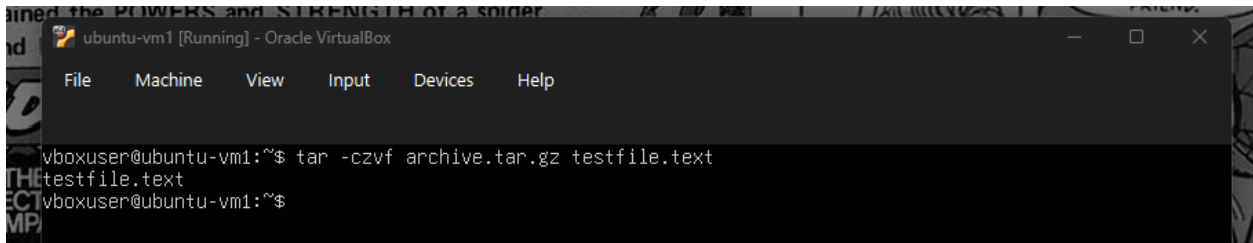


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ time sleep 2
real    0m2.009s
user    0m0.005s
sys     0m0.002s
vboxuser@ubuntu-vm1:~$ _
```

Waits measured time

2l

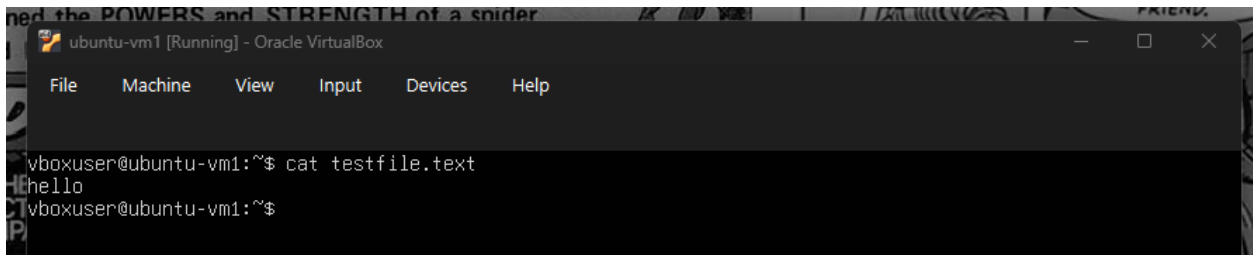


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ tar -czvf archive.tar.gz testfile.text
testfile.text
vboxuser@ubuntu-vm1:~$
```

Compresses files into an archive

2m

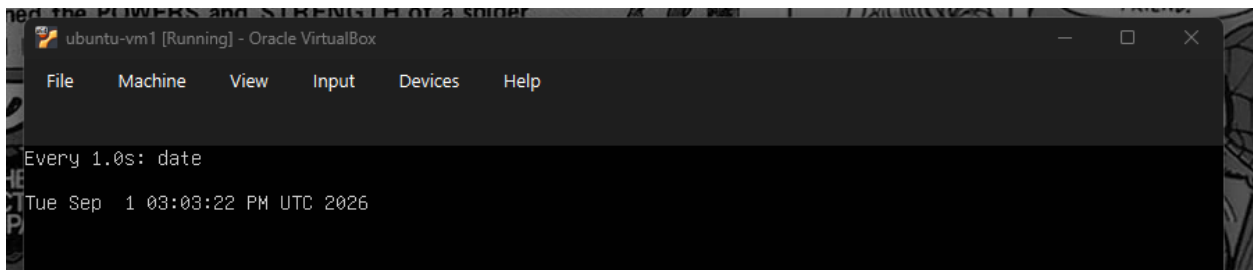


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ cat testfile.text
hello
vboxuser@ubuntu-vm1:~$
```

Prints file contents

2n



```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Every 1.0s: date
Tue Sep  1 03:03:22 PM UTC 2026
```

Display refreshing the display

2o

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ ps aux | head
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1   0.3   0.4 24680 16120 ?        Ss   14:39   0:04 /usr/lib/systemd/systemd --switched-root
root         2   0.0   0.0      0     0 ?        S    14:39   0:00 [kthreadd]
root         3   0.0   0.0      0     0 ?        S    14:39   0:00 [pool_workqueue_release]
root         4   0.0   0.0      0     0 ?        I<   14:39   0:00 [kworker/R-rcu_gp]
root         5   0.0   0.0      0     0 ?        I<   14:39   0:00 [kworker/R-sync_wq]
root         6   0.0   0.0      0     0 ?        I<   14:39   0:00 [kworker/R-kvfree_rcu_reclaim]
root         7   0.0   0.0      0     0 ?        I<   14:39   0:00 [kworker/R-slub_flushwq]
root         8   0.0   0.0      0     0 ?        I<   14:39   0:00 [kworker/R-netns]
root        12   0.0   0.0      0     0 ?        I    14:39   0:00 [kworker/u8:0-ipv6_addrconf]
vboxuser@ubuntu-vm1:~$
```

Lists running processes

2p

```
ed the POWERS and STRENGTH of a spider
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

top - 15:05:20 up 25 min, 1 user, load average: 0.01, 0.03, 0.06
Tasks: 115 total, 1 running, 114 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 99.9 id, 0.0 wa, 0.0 hi, 0.1 si, 0.0 st
MiB Mem : 3398.8 total, 2334.7 free, 463.7 used, 823.3 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 2935.1 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM    TIME+  COMMAND
    1 root        20   0   24680  16120 11732 S   0.0   0.5   0:04.89 systemd
    2 root        20   0     0     0     0 S   0.0   0.0   0:00.03 kthreadd
    3 root        20   0     0     0     0 S   0.0   0.0   0:00.00 pool_workqueue_release
    4 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-rcu_gp
    5 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-sync_wq
    6 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-kvfree_rcu_reclaim
    7 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-slub_flushwq
    8 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-netns
   12 root        20   0     0     0     0 I   0.0   0.0   0:00.00 kworker/u8:0-ipv6_addrconf
   13 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-mm_percpu_wq
   14 root        20   0     0     0     0 S   0.0   0.0   0:00.17 ksoftirqd/0
   15 root        20   0     0     0     0 I   0.0   0.0   0:00.35 rcu_preempt
   16 root        20   0     0     0     0 S   0.0   0.0   0:00.00 rcu_exp_par_gp_kthread_worker/0
   17 root        20   0     0     0     0 S   0.0   0.0   0:00.05 rcu_exp_gp_kthread_worker
   18 root        rt    0     0     0     0 S   0.0   0.0   0:00.05 migration/0
   19 root        20   0     0     0     0 S   0.0   0.0   0:00.00 kprobe-optimizer
   20 root       -51   0     0     0     0 S   0.0   0.0   0:00.00 idle_inject/0
   21 root        20   0     0     0     0 S   0.0   0.0   0:00.00 cpuhp/0
   22 root        20   0     0     0     0 S   0.0   0.0   0:00.00 cpuhp/1
   23 root       -51   0     0     0     0 S   0.0   0.0   0:00.00 idle_inject/1
   24 root        rt    0     0     0     0 S   0.0   0.0   0:00.04 migration/1
   25 root        20   0     0     0     0 S   0.0   0.0   0:01.04 ksoftirqd/1
   26 root        20   0     0     0     0 I   0.0   0.0   0:02.26 kworker/1:0-events
   28 root        20   0     0     0     0 I   0.0   0.0   0:00.92 kworker/u9:0-events_power_efficient
   29 root        20   0     0     0     0 I   0.0   0.0   0:00.12 kworker/u10:0-events_power_efficient
   30 root        20   0     0     0     0 S   0.0   0.0   0:00.02 kdevtmpfs
   31 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-inet_frag_wq
   32 root        20   0     0     0     0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   33 root        20   0     0     0     0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   34 root        20   0     0     0     0 S   0.0   0.0   0:00.00 kauditd
   35 root        20   0     0     0     0 S   0.0   0.0   0:00.00 khungtaskd
   36 root        20   0     0     0     0 S   0.0   0.0   0:00.00 oom_reaper
   38 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-writeback
   40 root        20   0     0     0     0 S   0.0   0.0   0:00.17 kcompactd0
   41 root        25   5     0     0     0 S   0.0   0.0   0:00.00 ksmd
   42 root        39  19     0     0     0 S   0.0   0.0   0:00.00 khugepaged
   43 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-kblockd
   44 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-blkcg_punt_bio
   45 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-kintegrityd
   46 root       -51   0     0     0     0 S   0.0   0.0   0:00.00 irq/9-acpi
   48 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-tpm_dev_wq
   49 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-ata_sff
   50 root         0 -20     0     0     0 I   0.0   0.0   0:00.00 kworker/R-md_bitmap
```

View of processes and resource usage

```
0.7% Tasks: 27, 37 thr, 86 kthr;
0.0% Load average: 0.00 0.02 0.06
241M/3.32G Uptime: 00:26:23
0K/0K

Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1 root 20 0 24680 16120 11732 S 0.0 0.5 0:04.89 /usr/lib/systemd/systemd --switched-root
437 root 19 -1 50656 18844 17508 S 0.0 0.5 0:00.35 /usr/lib/systemd/systemd-journald
478 root -11 0 272M 9644 7540 S 0.0 0.3 0:00.20 /usr/sbin/multipathd -d -s
486 root 20 0 272M 9644 0 S 0.0 0.3 0:00.00 /usr/sbin/multipathd -d -s
487 root -11 0 272M 9644 0 S 0.0 0.3 0:00.00 /usr/sbin/multipathd -d -s
488 root -11 0 272M 9644 0 S 0.0 0.3 0:00.00 /usr/sbin/multipathd -d -s
489 root -11 0 272M 9644 0 S 0.0 0.3 0:00.01 /usr/sbin/multipathd -d -s
490 root -11 0 272M 9644 0 S 0.0 0.3 0:00.51 /usr/sbin/multipathd -d -s
491 root -11 0 272M 9644 0 S 0.0 0.3 0:00.00 /usr/sbin/multipathd -d -s
492 root -11 0 272M 9644 0 S 0.0 0.3 0:00.00 /usr/sbin/multipathd -d -s
500 root 20 0 41248 15388 9224 S 0.0 0.4 0:00.58 /usr/lib/systemd/systemd-udev
505 systemd-re 20 0 22620 14736 12148 S 0.0 0.4 0:00.14 /usr/lib/systemd/systemd-resolved
749 systemd-ne 20 0 21432 12304 10732 S 0.0 0.4 0:00.18 /usr/lib/systemd/systemd-networkd
893 root 20 0 2888 1944 1824 S 0.0 0.1 0:00.11 /bin/sh /usr/lib/systemd/scripts/chronyd-
896 root 20 0 7060 3392 3152 S 0.0 0.1 0:00.04 /usr/sbin/cron -f -P
897 messagebus 20 0 9000 5704 4684 S 0.0 0.2 0:00.38 @dbus-daemon --system --address=systemd:
904 root 20 0 46728 30244 15100 S 0.0 0.9 0:00.76 /usr/bin/python3 /usr/bin/networkd-dispat
905 polkitd 20 0 299M 8428 7612 S 0.0 0.2 0:00.25 /usr/lib/polkit-1/polkitd --no-debug --lo
908 root 20 0 1841M 41984 28716 S 0.0 1.2 0:00.57 /usr/lib/snapd/snapd
911 root 20 0 18560 9700 8376 S 0.0 0.3 0:00.30 /usr/lib/systemd/systemd-logind
913 root 20 0 530M 15220 12836 S 0.0 0.4 0:00.36 /usr/libexec/udisks2/udisksd
945 root 20 0 10744 7768 6600 S 0.0 0.2 0:00.06 sshd: /usr/sbin/sshd -D [listener] 0 of 1
946 root 20 0 530M 15220 0 S 0.0 0.4 0:00.04 /usr/libexec/udisks2/udisksd
950 root 20 0 530M 15220 0 S 0.0 0.4 0:00.00 /usr/libexec/udisks2/udisksd
952 root 20 0 530M 15220 0 S 0.0 0.4 0:00.02 /usr/libexec/udisks2/udisksd
974 _chrony 20 0 20556 6364 5476 S 0.0 0.2 0:00.78 /usr/sbin/chronyd -n -F 1
982 syslog 20 0 215M 6096 4660 S 0.0 0.2 0:00.17 /usr/sbin/rsyslogd -n -iNONE
988 root 20 0 1841M 41984 0 S 0.0 1.2 0:00.30 /usr/lib/snapd/snapd
989 root 20 0 1841M 41984 0 S 0.0 1.2 0:00.00 /usr/lib/snapd/snapd
990 root 20 0 1841M 41984 0 S 0.0 1.2 0:00.07 /usr/lib/snapd/snapd
991 root 20 0 8760 5572 4708 S 0.0 0.2 0:00.10 login -- vboxuser
992 root 20 0 1841M 41984 0 S 0.0 1.2 0:00.00 /usr/lib/snapd/snapd
995 polkitd 20 0 299M 8428 0 S 0.0 0.2 0:00.05 /usr/lib/polkit-1/polkitd --no-debug --lo
997 polkitd 20 0 299M 8428 0 S 0.0 0.2 0:00.00 /usr/lib/polkit-1/polkitd --no-debug --lo
998 polkitd 20 0 299M 8428 0 S 0.0 0.2 0:00.02 /usr/lib/polkit-1/polkitd --no-debug --lo
1004 _chrony 20 0 12096 2448 1808 S 0.0 0.1 0:00.02 /usr/sbin/chronyd -n -F 1
1017 root 20 0 390M 14204 11944 S 0.0 0.4 0:00.34 /usr/sbin/ModemManager
1019 root 20 0 123M 32976 18356 S 0.0 0.9 0:00.38 /usr/bin/python3 /usr/share/unattended-up
1024 root 20 0 530M 15220 0 S 0.0 0.4 0:00.00 /usr/libexec/udisks2/udisksd
1025 root 20 0 530M 15220 0 S 0.0 0.4 0:00.00 /usr/libexec/udisks2/udisksd
1027 syslog 20 0 215M 6096 0 S 0.0 0.2 0:00.01 /usr/sbin/rsyslogd -n -iNONE

F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice F8Nice + F9Kill F10Quit
```

top view more detail

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ echo 'int main(){return 0;}' > test.c
vboxuser@ubuntu-vm1:~$ gcc test.c -o test
vboxuser@ubuntu-vm1:~$ ls -l test
-rwxrwxr-x 1 vboxuser vboxuser 15768 Sep  1 15:07 test
vboxuser@ubuntu-vm1:~$ ./test
vboxuser@ubuntu-vm1:~$ echo "Exit code : $?"
Exit code : 0
vboxuser@ubuntu-vm1:~$
```

Compiles c code

2s

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ tail /var/log/syslog
2026-09-01T14:50:30.673020+00:00 ubuntu-vm1 systemd[1]: Finished fwupd-refresh.service - Refresh fwupd metad
2026-09-01T14:54:45.332201+00:00 ubuntu-vm1 systemd[1]: Starting systemd-tmpfiles-clean.service - Cleanup of
2026-09-01T14:54:45.478215+00:00 ubuntu-vm1 systemd[1]: systemd-tmpfiles-clean.service: Deactivated successf
2026-09-01T14:54:45.478681+00:00 ubuntu-vm1 systemd[1]: Finished systemd-tmpfiles-clean.service - Cleanup of
2026-09-01T15:00:28.811449+00:00 ubuntu-vm1 systemd[1]: Starting sysstat-collect.service - system activity a
2026-09-01T15:00:28.929061+00:00 ubuntu-vm1 systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-09-01T15:00:28.931049+00:00 ubuntu-vm1 systemd[1]: Finished sysstat-collect.service - system activity a
2026-09-01T15:10:18.972749+00:00 ubuntu-vm1 systemd[1]: Starting sysstat-collect.service - system activity a
2026-09-01T15:10:19.062230+00:00 ubuntu-vm1 systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-09-01T15:10:19.062440+00:00 ubuntu-vm1 systemd[1]: Finished sysstat-collect.service - system activity a
vboxuser@ubuntu-vm1:~$
```

Shows the last lines of a file

2t

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
vboxuser@ubuntu-vm1:~$ _
```

Searches text for a pattern

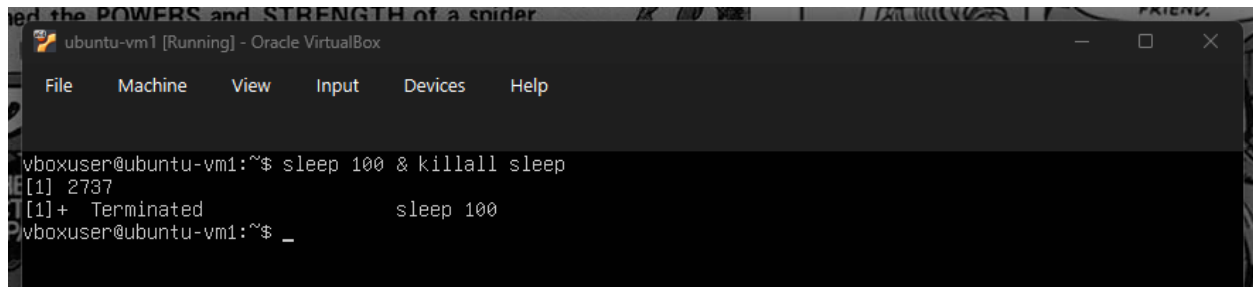
2u

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ sleep 100 & kill %1
[1] 2717
[1]+  Terminated                  sleep 100
vboxuser@ubuntu-vm1:~$ _
```

Kills a process

2v

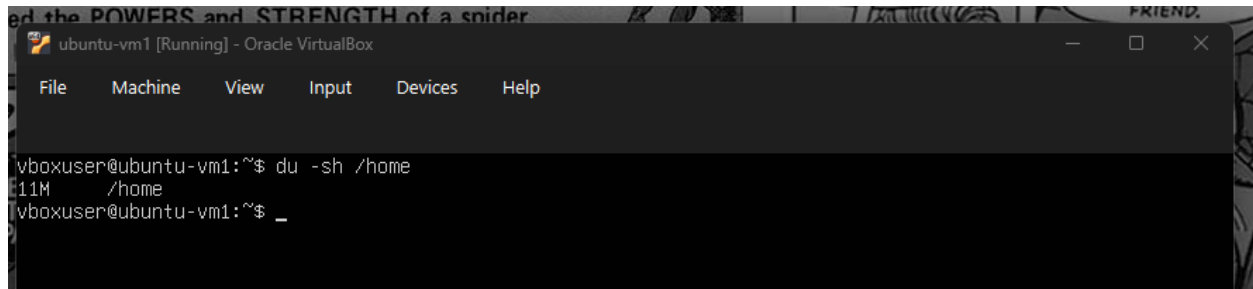


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ sleep 100 & killall sleep
[1] 2737
[1]+  Terminated                  sleep 100
vboxuser@ubuntu-vm1:~$ _
```

Kills processes by name

2w

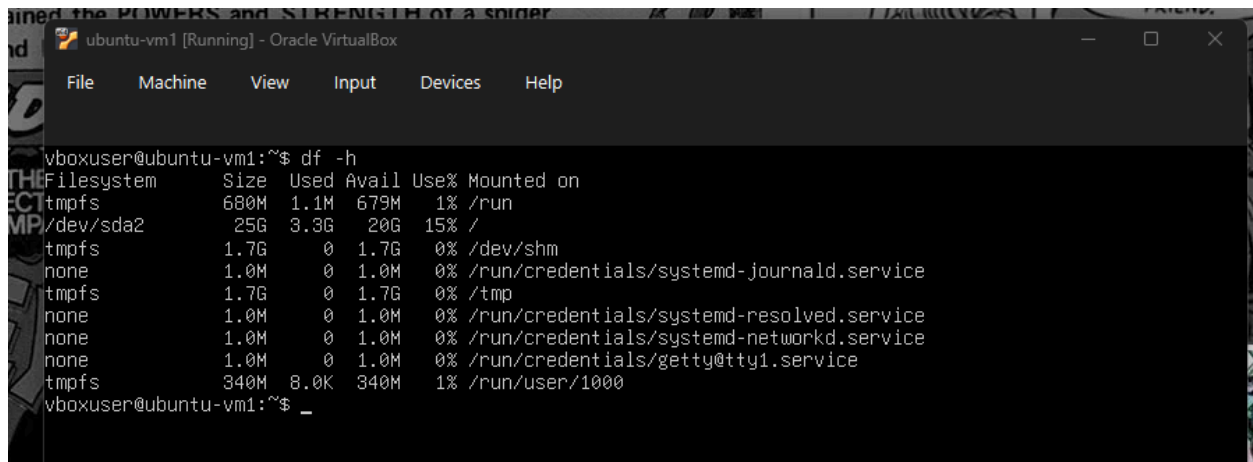


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ du -sh /home
11M    /home
vboxuser@ubuntu-vm1:~$ _
```

Shows disk usage of files/directories

2x

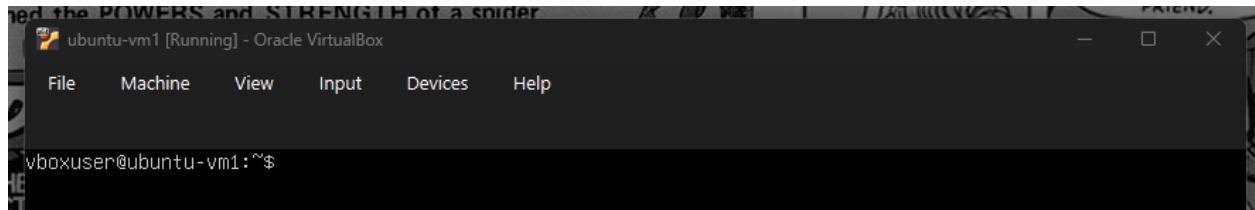
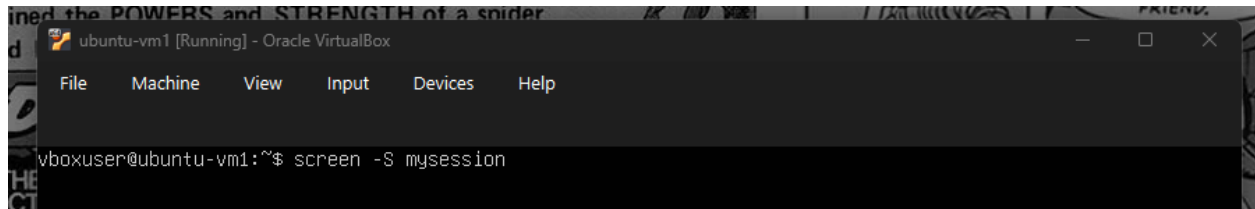


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           680M  1.1M  679M   1% /run
/dev/sda2       25G   3.3G   20G  15% /
tmpfs           1.7G   0  1.7G   0% /dev/shm
none            1.0M   0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.7G   0  1.7G   0% /tmp
none            1.0M   0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M   0  1.0M   0% /run/credentials/systemd-networkd.service
none            1.0M   0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           340M   8.0K  340M   1% /run/user/1000
vboxuser@ubuntu-vm1:~$ _
```

Shows disk space per filesystem

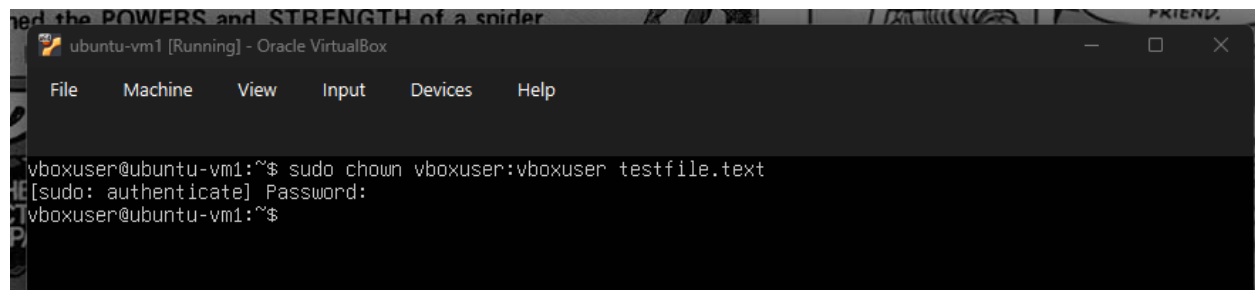
2y



Persistent terminal session

2z

2bb

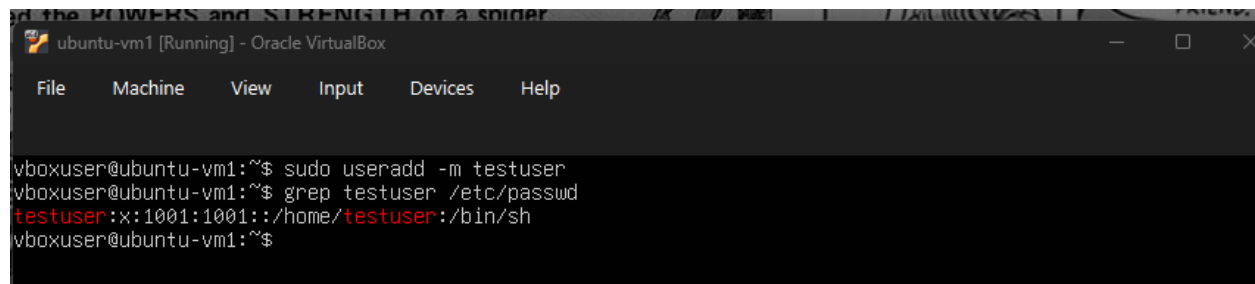


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ sudo chown vboxuser:vboxuser testfile.text
[sudo: authenticate] Password:
vboxuser@ubuntu-vm1:~$
```

Changes the file owner/group

2cc



```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ sudo useradd -m testuser
vboxuser@ubuntu-vm1:~$ grep testuser /etc/passwd
testuser:x:1001:1001::/home/testuser:/bin/sh
vboxuser@ubuntu-vm1:~$
```

2dd

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

ls(1) General Commands Manual

NAME
ls - List directory contents. Ignore files and directories starting with a '.' by default

SYNOPSIS
ls [--help] [--format] [-C ] [-l|--long] [-x ] [-T|--tabsize] [-m ] [--zero] [-D|--dired] [--quoting-style] [-N|--literal] [-b|--escape] [-Q|--quote-name] [-q|--hide-control-chars] [--show] [-I|--ignore] [-B|--ignore-backups] [--sort] [-S ] [-t ] [-v ] [-X ] [-U ] [-L|--dereference] [--dereference-command-line] [-G|--no-group] [--author] [-a|--all] [-A|--almost-all] [-f ] [-d|--directo] [--block-size] [-i|--inode] [-r|--reverse] [-R|--recursive] [-w|--width] [-s|--size] [--color] [--ind] [--time-style] [--full-time] [-Z|--context] [--group-directories-first] [-V|--version] [paths]

DESCRIPTION
List directory contents. Ignore files and directories starting with a '.' by default

OPTIONS
--help Print help information.

--format
Set the display format.

-C Display the files in columns.

-l, --long
Display detailed information.

-x List entries in rows instead of in columns.

-T, --tabsize <COLS>
Assume tab stops at each COLS instead of 8

-m List entries separated by commas.

--zero List entries separated by ASCII NUL characters.

-D, --dired
generate output designed for Emacs' dired (Directory Editor) mode

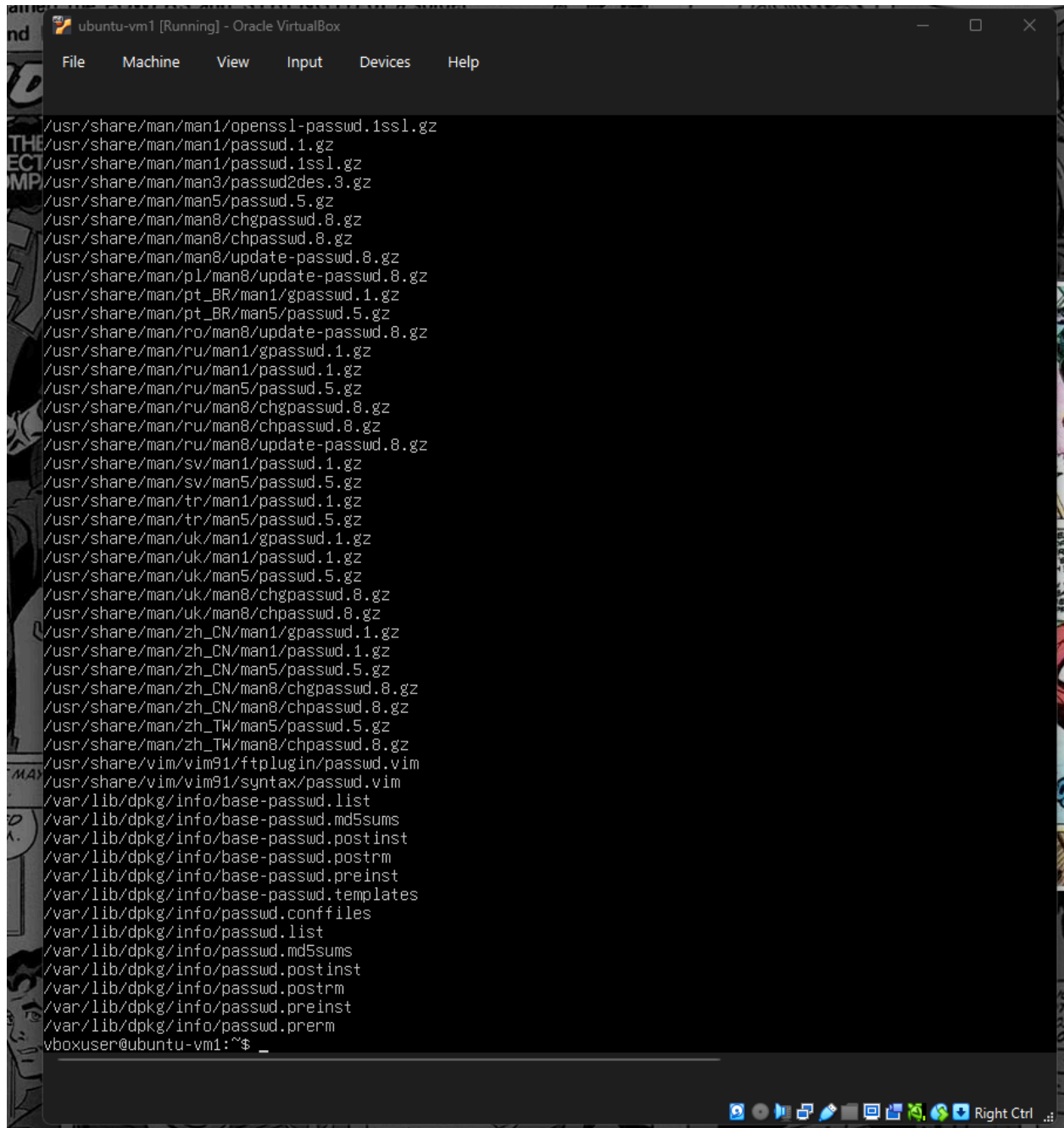
--hyperlink[=<WHEN>] [default: never]
hyperlink file names WHEN

Possible values:
• always
• auto
• never

Manual page ls(1) line 1 (press h for help or q to quit)
```

Command manual page

2ee

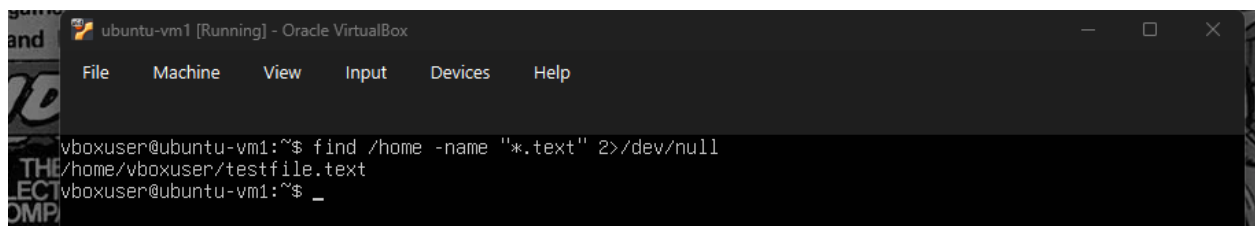


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

/usr/share/man/man1/openssl-passwd.1ssl.gz
/usr/share/man/man1/passwd.1.gz
/usr/share/man/man1/passwd.1ssl.gz
/usr/share/man/man3/passwd2des.3.gz
/usr/share/man/man5/passwd.5.gz
/usr/share/man/man8/chgpasswd.8.gz
/usr/share/man/man8/chpasswd.8.gz
/usr/share/man/man8/update-passwd.8.gz
/usr/share/man/pl/man8/update-passwd.8.gz
/usr/share/man/pt_BR/man1/gpasswd.1.gz
/usr/share/man/pt_BR/man5/passwd.5.gz
/usr/share/man/ro/man8/update-passwd.8.gz
/usr/share/man/ru/man1/gpasswd.1.gz
/usr/share/man/ru/man1/passwd.1.gz
/usr/share/man/ru/man5/passwd.5.gz
/usr/share/man/ru/man8/chgpasswd.8.gz
/usr/share/man/ru/man8/chpasswd.8.gz
/usr/share/man/ru/man8/update-passwd.8.gz
/usr/share/man/sv/man1/passwd.1.gz
/usr/share/man/sv/man5/passwd.5.gz
/usr/share/man/tr/man1/passwd.1.gz
/usr/share/man/tr/man5/passwd.5.gz
/usr/share/man/uk/man1/gpasswd.1.gz
/usr/share/man/uk/man1/passwd.1.gz
/usr/share/man/uk/man5/passwd.5.gz
/usr/share/man/uk/man8/chgpasswd.8.gz
/usr/share/man/uk/man8/chpasswd.8.gz
/usr/share/man/zh_CN/man1/gpasswd.1.gz
/usr/share/man/zh_CN/man1/passwd.1.gz
/usr/share/man/zh_CN/man5/passwd.5.gz
/usr/share/man/zh_CN/man8/chgpasswd.8.gz
/usr/share/man/zh_CN/man8/chpasswd.8.gz
/usr/share/man/zh_TW/man5/passwd.5.gz
/usr/share/man/zh_TW/man8/chpasswd.8.gz
/usr/share/vim/vim91/ftplugin/passwd.vim
/usr/share/vim/vim91/syntax/passwd.vim
/var/lib/dpkg/info/base-passwd.list
/var/lib/dpkg/info/base-passwd.md5sums
/var/lib/dpkg/info/base-passwd.postinst
/var/lib/dpkg/info/base-passwd.postrm
/var/lib/dpkg/info/base-passwd.preinst
/var/lib/dpkg/info/base-passwd.templates
/var/lib/dpkg/info/passwd.conf files
/var/lib/dpkg/info/passwd.list
/var/lib/dpkg/info/passwd.md5sums
/var/lib/dpkg/info/passwd.postinst
/var/lib/dpkg/info/passwd.postrm
/var/lib/dpkg/info/passwd.preinst
/var/lib/dpkg/info/passwd.prerm
vboxuser@ubuntu-vm1:~$ _
```

Finds files by name via prebuilt index

2ff

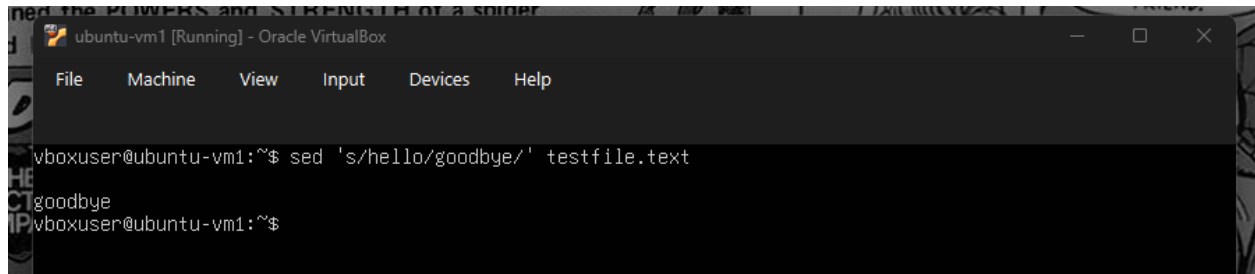


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ find /home -name "*.text" 2>/dev/null
/home/vboxuser/testfile.text
vboxuser@ubuntu-vm1:~$ _
```

Searches filesystem

2gg

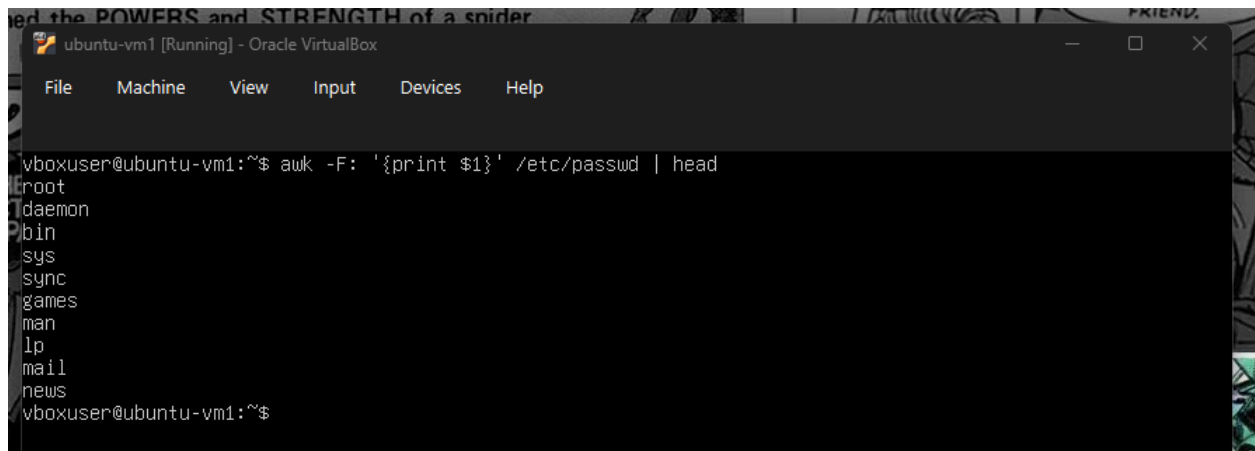


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ sed 's/hello/goodbye/' testfile.txt
goodbye
vboxuser@ubuntu-vm1:~$
```

Finds and replaces text in stream

2hh

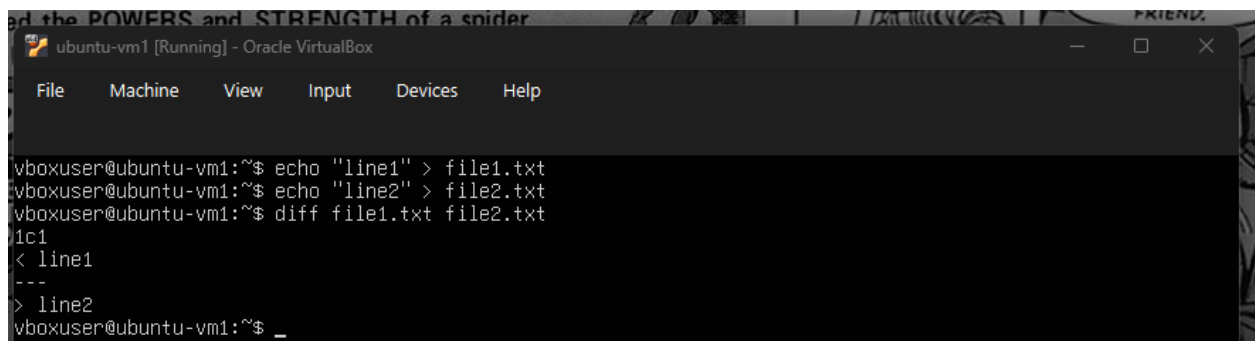


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ awk -F: '{print $1}' /etc/passwd | head
root
daemon
bin
sys
sync
games
man
lp
mail
news
vboxuser@ubuntu-vm1:~$
```

column wise text process

2ii

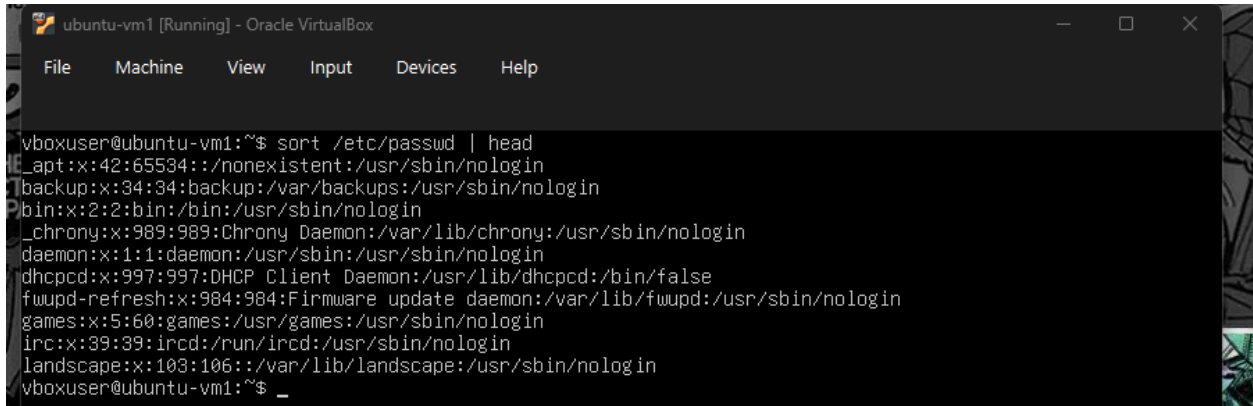


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ echo "line1" > file1.txt
vboxuser@ubuntu-vm1:~$ echo "line2" > file2.txt
vboxuser@ubuntu-vm1:~$ diff file1.txt file2.txt
1c1
< line1
---
> line2
vboxuser@ubuntu-vm1:~$ _
```

Compares 2 files

2jj

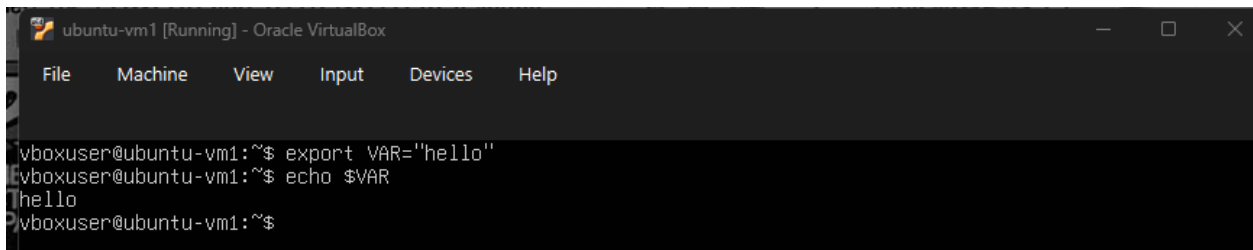


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ sort /etc/passwd | head
_apt:x:42:65534::/nonexistent:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
_chrony:x:989:989:Chrony Daemon:/var/lib/chrony:/usr/sbin/nologin
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
dhcpd:x:997:997:DHCP Client Daemon:/usr/lib/dhcpd:/bin/false
fwupd-refresh:x:984:984:Firmware update daemon:/var/lib/fwupd:/usr/sbin/nologin
games:x:5:60:games:/usr/games:/usr/sbin/nologin
irc:x:39:39:ircd:/run/ircd:/usr/sbin/nologin
landscape:x:103:106::/var/lib/landscape:/usr/sbin/nologin
vboxuser@ubuntu-vm1:~$ _
```

Sorts lines of text

2kk

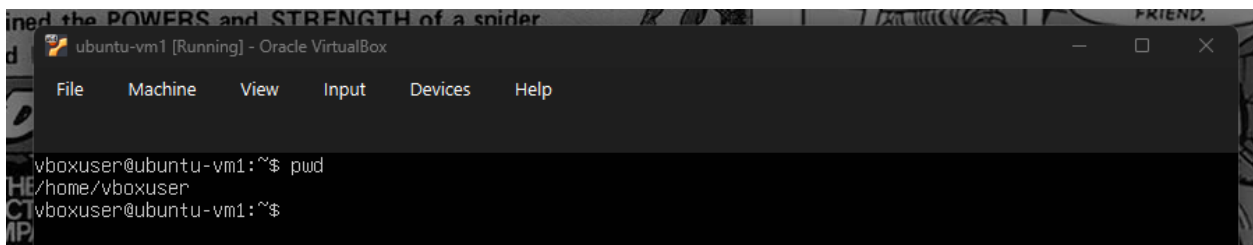


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ export VAR="hello"
vboxuser@ubuntu-vm1:~$ echo $VAR
hello
vboxuser@ubuntu-vm1:~$
```

Sets an environment variable

2ll

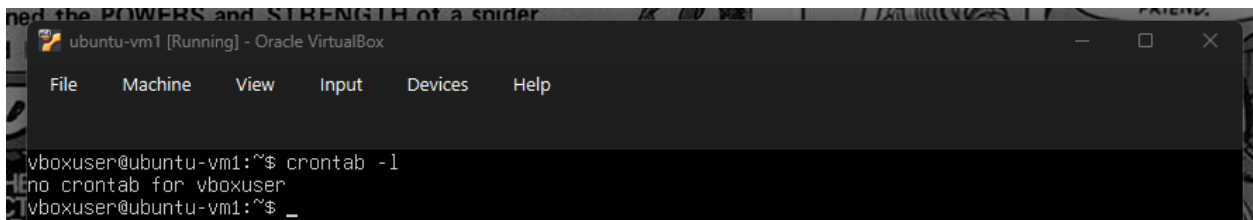


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ pwd
/home/vboxuser
vboxuser@ubuntu-vm1:~$
```

Prints current working directory

2mm

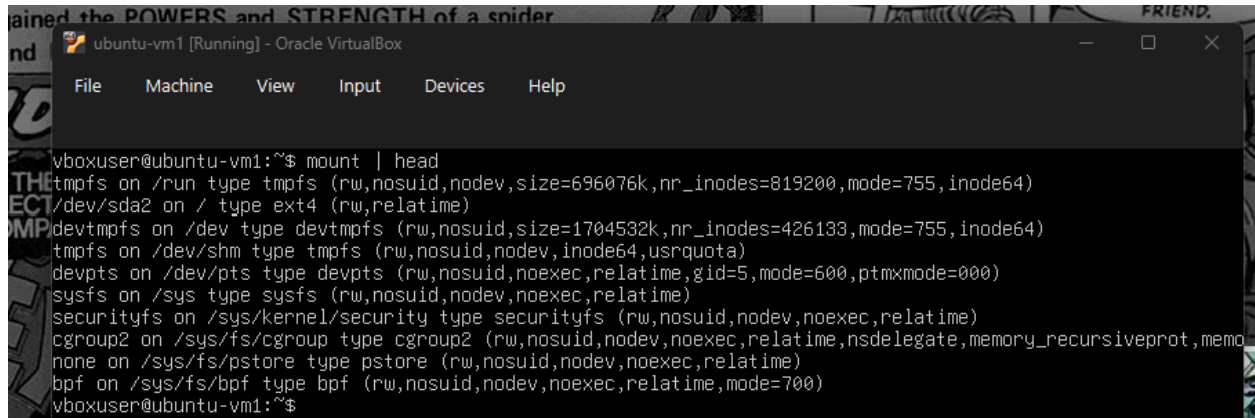


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ crontab -l
no crontab for vboxuser
vboxuser@ubuntu-vm1:~$ _
```

Shows schedule of recurring tasks

2nn

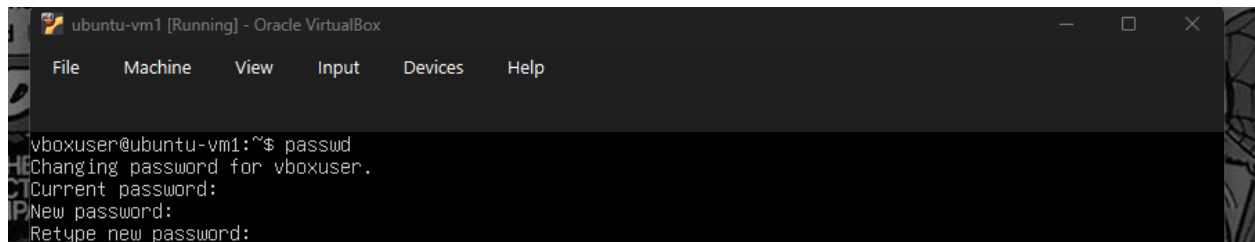


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ mount | head
tmpfs on /run type tmpfs (rw,nosuid,nodev,size=696076k,nr_inodes=819200,mode=755,inode64)
/dev/sda2 on / type ext4 (rw,relatime)
devtmpfs on /dev type devtmpfs (rw,nosuid,size=1704532k,nr_inodes=426133,mode=755,inode64)
tmpfs on /dev/shm type tmpfs (rw,nosuid,nodev,inode64,usrquota)
devpts on /dev/pts type devpts (rw,nosuid,noexec,relatime,gid=5,mode=600,ptmxmode=000)
sysfs on /sys type sysfs (rw,nosuid,nodev,noexec,relatime)
securityfs on /sys/kernel/security type securityfs (rw,nosuid,nodev,noexec,relatime)
cgroup2 on /sys/fs/cgroup type cgroup2 (rw,nosuid,nodev,noexec,relatime,nsdelegate,memory_recursiveprot,memo
none on /sys/fs/pstore type pstore (rw,nosuid,nodev,noexec,relatime)
bpf on /sys/fs/bpf type bpf (rw,nosuid,nodev,noexec,relatime,mode=700)
vboxuser@ubuntu-vm1:~$
```

Shows attached filesystems

2oo

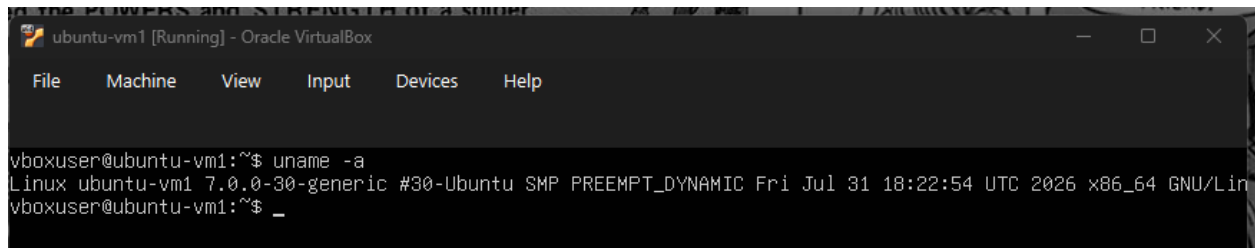


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ passwd
Changing password for vboxuser.
Current password:
New password:
Retype new password:
```

Changes a users password

2pp

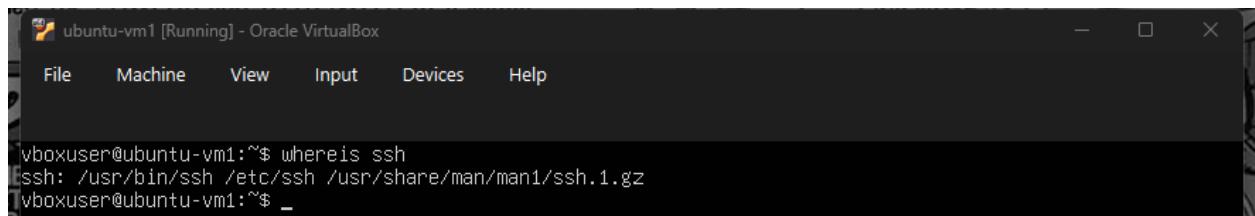


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ uname -a
Linux ubuntu-vm1 7.0.0-30-generic #30-Ubuntu SMP PREEMPT_DYNAMIC Fri Jul 31 18:22:54 UTC 2026 x86_64 GNU/Linux
vboxuser@ubuntu-vm1:~$ _
```

Prints system info

2qq

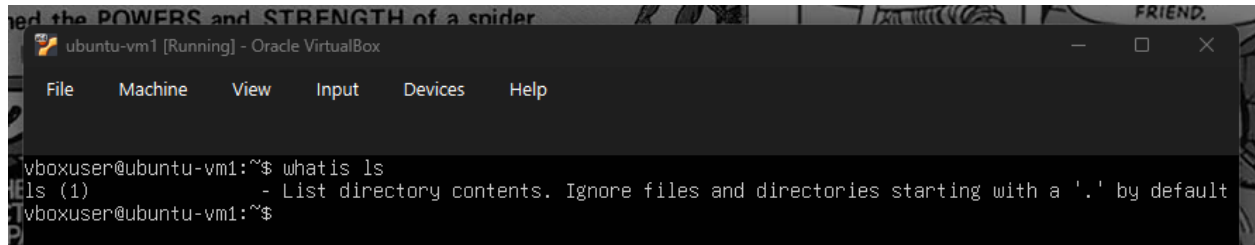


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ whereis ssh
ssh: /usr/bin/ssh /etc/ssh /usr/share/man/man1/ssh.1.gz
vboxuser@ubuntu-vm1:~$ _
```

Find source location for a command

2rr

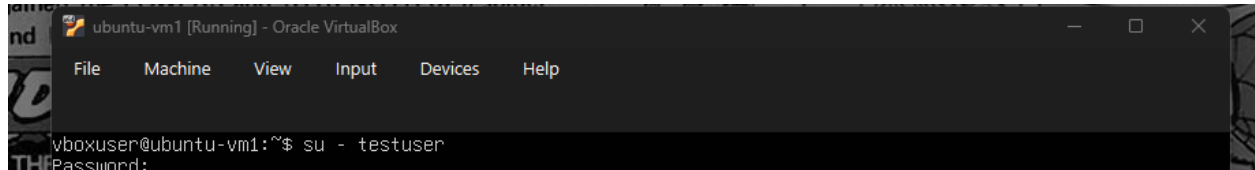


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ whatis ls
ls (1)             - List directory contents. Ignore files and directories starting with a '.' by default
vboxuser@ubuntu-vm1:~$
```

Description of a command

2ss

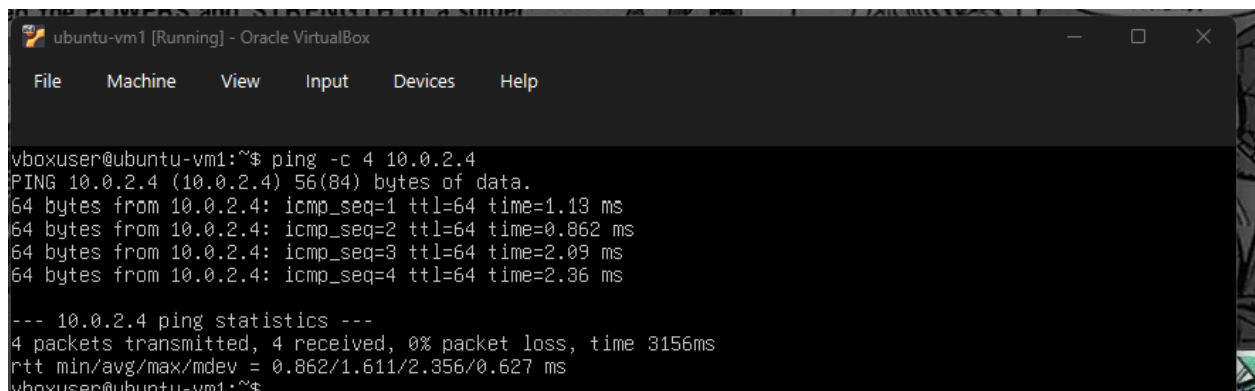


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ su - testuser
Password:
```

Switches to another user account

2tt



```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ ping -c 4 10.0.2.4
PING 10.0.2.4 (10.0.2.4) 56(84) bytes of data:
64 bytes from 10.0.2.4: icmp_seq=1 ttl=64 time=1.13 ms
64 bytes from 10.0.2.4: icmp_seq=2 ttl=64 time=0.862 ms
64 bytes from 10.0.2.4: icmp_seq=3 ttl=64 time=2.09 ms
64 bytes from 10.0.2.4: icmp_seq=4 ttl=64 time=2.36 ms

--- 10.0.2.4 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3156ms
rtt min/avg/max/mdev = 0.862/1.611/2.356/0.627 ms
vboxuser@ubuntu-vm1:~$
```

Test network connection

2uu


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ traceroute 8.8.8.8
traceroute to 8.8.8.8 (8.8.8.8), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.1)  0.774 ms  0.638 ms  0.574 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Shows network path to a destination

2vv

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ date
Tue Sep  1 03:43:29 PM UTC 2026
vboxuser@ubuntu-vm1:~$
```

Shows current date/time

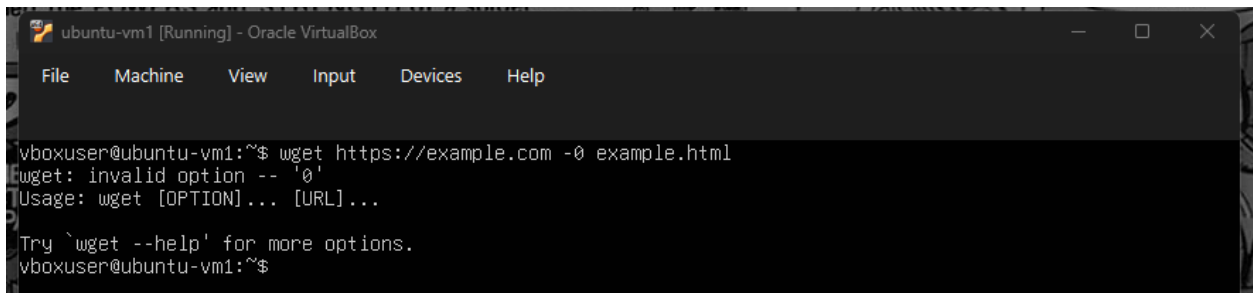
2ww

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ time ls
archive.tar.gz file1.txt file2.txt test test.c testfile.img testfile.text
real    0m0.009s
user    0m0.008s
sys     0m0.001s
vboxuser@ubuntu-vm1:~$
```

This is a repeat of k

2xx



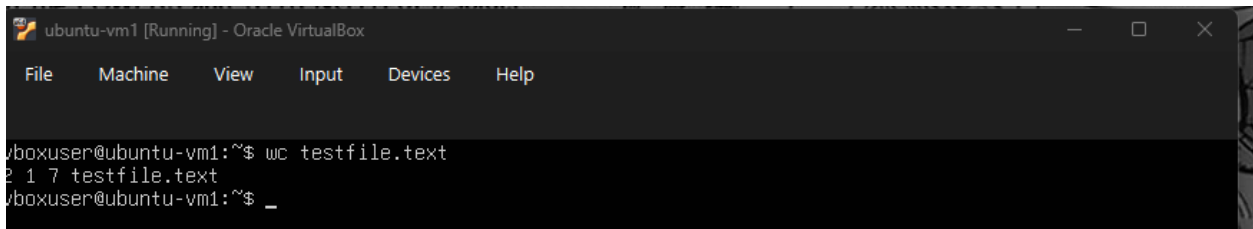
```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ wget https://example.com -O example.html
wget: invalid option -- 'O'
Usage: wget [OPTION]... [URL]...

Try `wget --help' for more options.
vboxuser@ubuntu-vm1:~$
```

Downloads files from the internet

2yy

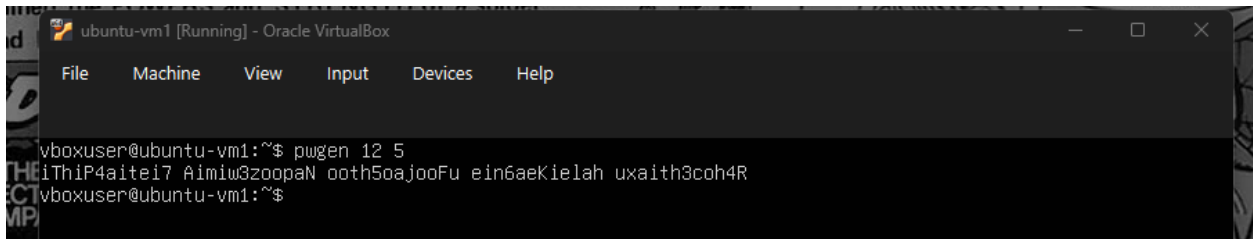


```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ wc testfile.txt
2 1 7 testfile.txt
vboxuser@ubuntu-vm1:~$ _
```

Counts lines words characters

2zz



```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ pwgen 12 5
iThiP4aitei7 AimiW3zoopaN ooth5oaJooFu ein6aeKielah uxaith3coh4R
vboxuser@ubuntu-vm1:~$
```

Generates random password

3a

```
ubuntu-vm1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

#!/bin/bash
if [ $# -ne 2 ] ; then
    echo "Usage: $0 <filename> <num_records>"
    exit 1
fi

FILENAME="$1"
NUM_RECORDS="$2"
NPROC=$((NPROC))
CHUNK=$(( (NUM_RECORDS + NPROC - 1) / NPROC ))
TMPDIR=$(mktemp -d)

gen_chunk(){
    awk -v n="$1" -v seed="$2" '
    BEGIN{
        srand(seed);
        chars = "ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789";
        clen = length(chars);
        for(i = 0; i < n; i++){
            int1 = int(rand() * 4294967296);
            int2 = int(rand() * 4294967296);
            str = "";
            for(j = 0; j < 100; j++){
                str = str substr(chars, int(rand() * clen) + 1, 1);
            }
            print int1,int2,str;
        }
    }' "$@"
}

for i in $(seq 1 $NPROC); do
    SEEDVAL=$((RANDOM * i + $RANDOM))
    gen_chunk "$CHUNK" "$TMPDIR/part_$(i).txt" "$SEEDVAL" &
done
wait

cat "$TMPDIR"/part_*.txt | head -n "$NUM_RECORDS" > "$FILENAME"
rm -rf "$TMPDIR"

"generate-dataset.sh" 40L, 827B
34,60-61 All
Right Ctrl .it
```

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help

vboxuser@ubuntu-vm1:~$ ./generate-dataset.sh test_debug6.txt 10
vboxuser@ubuntu-vm1:~$ cat test_debug6.txt
974365454 423871525 6a1e0nerpqbgJrae1TLMySS14SAvp3qHo5T22t1nYNGSVBSE4oXZDwb62mVYIU2kLQzTJFmJPu8MY57KS9bBHpJhDR6tUEfx41v
2584772132 657750096 P7B1ubYrmReql7MfacS0a39kKD1gKxUs6610BwrfXE2K1BEt461UNb05hmQ69c3a0gesY1Rgz19G0gVKWfn9MnDXuRdtZVratuU
2527664880 4049618497 20TfArJ009P3FLxLIuug5x8LbL5OmZ0UtsfshgHpeerHuWnUKL7PC1zkeRNSegZ89vEXSrTbMm1CK41nsh5U0oF61a8qu010r
2447911881 3819443101 gH9000p2IL0nmEf1o29PgaYEiUxSVZ2nfm1av9JJMSL3RwclnZDOnBmW60zVT10vN5MV0ZK62ClTpol1KE3NKbvJUKt3kVGrS
1010963792 1067331954 J1V00f20uIgIng0aFBNiKJF7J9tXy5t0eNW081btFkoEoZK4pr562TscdXSScfqHo4tT1govBsm0N1Ag3ALXj7aU6cCPV42X1pe
3642185419 2530871637 HsGQc97zCysHf0Qjlpkb6gSkqrBQU42j0Fa00dhAmVYGHcEghSL8cefUp12Td4W00Q6LIqysFvDe0hOGe4dskJ66EotHrToFQud9
944563957 180523640 m0GFe5IUT568f1Nv120gkarXhPvmamQx0f1PrbR0UsDVmIXUZjzkZmw1AW7uL1vyvshGB34f0H0JNNT20smmp2yuvwU0JWgEMQJ
3024199222 2270891361 1LzAIUyPnQ4UBpAHNbM0rIu19A2PeH1gyJ2AJKU31mXbtQp19p0aFV91kMRm1fekSW0AnNLEu2z8gpFrEKUk8YamTS4oyJLfY10q
1201301961 1395609752 J1CQCFRf6J8JxkqHbFvXUSA0PcPgauCnryV79yBvaEg99aU61qTLTv69Ks20CBW3qG6BQcE1qS3jhSa6qHhTVk0EA30fdTbmaQR
4192333679 2692043479 DYyGg4cHHu0kuECrLA2zpImD0pcvniX02hB0mx0SA4pJPrQKxvNPKAoX4DU0sV0u6m8CaLLemoXPInfk0qY2RBGJioVkuLL72tZ8
vboxuser@ubuntu-vm1:~$ sort test_debug6.txt | uniq -d
vboxuser@ubuntu-vm1:~$ awk '{print NF}' test_debug6.txt
> ^C
vboxuser@ubuntu-vm1:~$ awk '{print NF}' test_debug6.txt
3
3
3
3
3
3
3
3
3
3
3
3
vboxuser@ubuntu-vm1:~$ _
```

```
ubuntu-vm1 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
vboxuser@ubuntu-vm1:~$ time ./generate-dataset.sh test10m.txt 10000000
[13777.972012] rcu: INFO: rcu_preempt self-detected stall on CPU
[13777.972074] rcu: #0-...!: (1 GPs behind) idle=c09c/0/0x1 softirq=38178/38179 fqs=0
[13777.972110] rcu: #t=97001 jiffies g=70077 q=35 ncpu=2
[13777.972129] rcu: rcu_preempt kthread timer wakeup didn't happen for 96998 jiffies! g70077 f0x0 RCU_GP_WAIT_FQS(5) ->state=0x402
[13777.972144] rcu: #Possible timer handling issue on cpu=1 timer-softirq=29976
[13777.972177] rcu: rcu_preempt kthread starved for 97001 jiffies! g70077 f0x0 RCU_GP_WAIT_FQS(5) ->state=0x402 ->cpu=1
[13777.972202] rcu: #Unless rcu_preempt kthread gets sufficient CPU time, OOM is now expected behavior.
[13777.972226] rcu: RCU grace-period kthread stack dump:
[13777.972710] rcu: Stack dump where RCU GP kthread last ran:
[13787.984837] nmi_backtrace_stall_check: CPU 1: NMIs are not reaching exc_nmi() handler (CPU was never in an NMI handler function)
[13787.984895] nmi_backtrace_stall_check: last activity: 4308539014 jiffies ago.

real    1m51.955s
user    2m25.411s
sys      0m52.283s
vboxuser@ubuntu-vm1:~$ _
```

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
vboxuser@ubuntu-vm1:~$ wc -l test10m.txt
1000000 test10m.txt
vboxuser@ubuntu-vm1:~$ head -3 test10m.txt
258590277 3258699821 HHp6WxC36uRcb3vHx7FFpYVJfgoQNG6Bxv4iWg0y3p2eAb8FVF200huMD1YV32vq31xEgDxg9pgQKvcXu5dsae546UWBSWDhRFZ2
3990519159 857190935 3apIyLLAI20gSPHXJkFp3Zct0byXXJAn239zKhp751nnJSfWEMg3Rg1Ac0dhWQUSsiKdcvXoI220h1S0uYEJpJ8hvcOrF96nJ2W
927404690 2886555349 cWj0Mvpf0s4n9ex4av3mTuAYpR13J0dMKLN0V2chN00bJaToE107kLHzb1r9hQoMdqmoE51vwgu1RL2kvcFTY2R5Yg80uqvSNIFq
vboxuser@ubuntu-vm1:~$ tail -3 test10m.txt
2142250247 3950699277 J51imRtQ97J0pkdKLOBhJ4YeC0HqfVB29LC1t2n1rTZCJm6XNsDHY10Hy7ghx46uSsECYK7ISxGtS2owLk61IyJqC2v5v50J4mbF
814481714 1376119060 lsgkeMmo1NRHY211JeHs7MYRczkd0w4I00Tyf6IQ8baJkceEuwunEbIh0NACuIM1VvTJZATWfIa8sT6hav8vY6adQCKQKxaS1EvJ
2645265046 3464885058 7HScqJF1040yR98uobcV4n3V1HvYfgE2K3ufw8InyEDs8gJnNJlpj10wUIXtdmVvyAf3zexpuujF1aC3BP98K0fCkerINi6a6pr
vboxuser@ubuntu-vm1:~$ awk '{print NF}' test10m.txt | sort | uniq -c
1000000 3
vboxuser@ubuntu-vm1:~$ _
```

3b

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
#!/bin/bash
if [ $# -lt 1 ]; then
    ech "Usage: $0 <input_file> [output_file]"
    exit 1
fi

INPUT_FILE="$1"
OUTPUT_FILE="${2:-${INPUT_FILE}.sorted}"
NPROC=$(nproc)

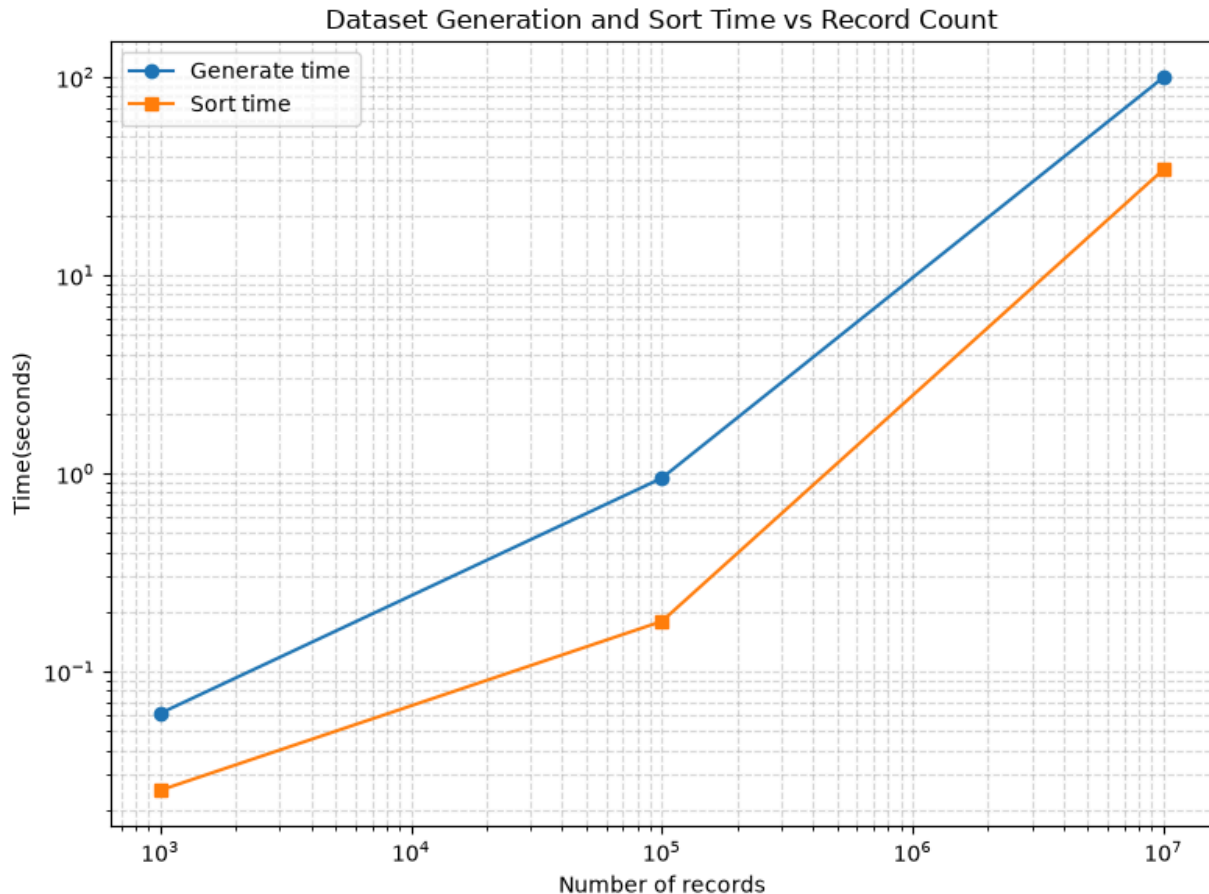
sort -n -k1,1 --parallel=$NPROC -S 50% "$INPUT_FILE" > "$OUTPUT_FILE"
echo "Sorted $INPUT_FILE -> $OUTPUT_FILE"
```

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
vboxuser@ubuntu-vm1:~$ time ./sort-data.sh test10m.txt
Sorted test10m.txt -> test10m.txt.sorted

real    0m28.101s
user    0m9.210s
sys     0m25.507s
vboxuser@ubuntu-vm1:~$ head -5 test10m.txt.sorted
of
340 2026916354 BZ1mb5kYPcPebx2nEiGSz16uuqHFVE4RsaA1JQykUrsTSsCjJtUQp9QzYQcwCnmwvCR4ghDbonv3X016L1JkLZV5hxF6hqJHUp
539 471973050 s8MzAUmkRgg1uQsN550SeXlwgHLnKgYd19rgtaLA8ByTa8XR16Edg8gKYsGsgRABQaEmfLyDYfcwIkLSLCx5pXYVc0zRfTsv06
672 1809592658 rC1cAw1jHe9bUz1gDMSGVpDLeNjrvguVDNzcM8hhvntX6XVGP4eYX08ad2u26ordk1akZ2Hp7BxqWAA0I0A5CBREfNqkxmMpFSoe
1855 11205021 UnJJBSzOS8Ndda27ReaUadE31HFu89094xFsa7uJRNSrGwFJKF2QaaH2285gadeHgCUj8C0U00XGpsPD9CACFqdsJ44Iop6Gt2a2
vboxuser@ubuntu-vm1:~$ awk '{print NF}' test10m.txt.sorted | sort | uniq -c
      1 1
10000000 3
vboxuser@ubuntu-vm1:~$
```

3c

```
ubuntu-vm1 [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 #!/bin/bash
2 SCALES="1000 100000 10000000"
3 RESULTS="benchmark_results.csv"
4
5 echo "records,generate_time_sec, sort_time_sec" > "$RESULTS"
6
7 for n in $SCALES; do
8     FILE="data_${n}.txt"
9     SORTED="data_${n}.txt.sorted"
10
11     GEN_START=$(date +%s.%N)
12     ./generate-dataset.sh "$FILE" "$n"
13     GEN_END=$(date +%s.%N)
14     GEN_TIME=$(echo "GEN_END - $GEN_START" | bc)
15
16     ./sort-data.sh "$FILE"
17     SORT_END=$(date +%s.%N)
18     SORT_TIME=$(echo "%SORT_END-%SORT_START" | bc)
19
20     echo "$n,$GEN_TIME,$SORT_TIME" >> "$RESULTS"
21     echo "n=$n generate=$GEN_TIMEs sort=$SORT_TIMEs"
22
23 done
24
25 echo "Benchmark complete. Results in $RESULTS"
26
27
```

4a

Processor settings is the number of physical CPU cores given to the guest OS as virtual CPUs. The guest OS has its own processors and threads across the virtual CPUs. The more virtual CPUs that can be run the more work in parallel. A lightweight VM running a single service that's not very CPU intensive like a small file server would be an example of a minimum scenario. A VM running CPU intensive workloads in parallel like the generated dataset would be an example of the maximum scenario. This is bad for a couple of reasons because if you could cause oversubscription, that being if you assign more vCPUs than your host has physical core, additionally giving a VM all your cores can leave your host OS with nothing making it slow.

4b

None: Guest runs with no paravirtualization interface, the hypervisor completely emulates hardware, and the OS will not know which more is virtualized. More compatible but slower, since every hardware interaction will go through full emulation

Legacy: This is an older version of paravirtualization interface kept for backward compatibility with older host/guest OS combinations.

Minimal: A minimal interface usually used for Mac users, this hides some of the virtualization signatures.

Hyper-V: Exposes the hyper v compatible paravirtualization interface to the guest letting the guest OS use fast path. Best for Windows users, since Windows has a built-in hyper v interface.

KVM: Exposes the KVM compatible interface, letting linux guests have faster access to clock synchronization, interrupt handling, and I/O

4c

IDE: This is integrated drive electronics. The oldest, slowest, limited to a small number of devices and slower transfer speeds. This would be best for a windows 98 or xp machine that doesn't have sata or ahci drivers.

SATA: Serial ATA. Modern, faster interface than IDE, supports more devices and has better throughput. A scenario for this would be a general purpose modern OS install, its a default between balance of compatibility and performance for everyday virtual disks.

NVME: Non volatile memory express. This is designed for solid data storage, communicates directly over the PCIe rather than through legacy storage protocols and offers more throughput with lower latency than SATA. An example of this is a high performance workload that is designed for I/O bound, a database server handling lots of read/writes or benchmarking its own storage performance.

4d

NAT: The vm shares the host's network connection through address translation, the vm goes outbound to the intern, but the outside world can't create connection into it. It's hidden behind the host like a router. A scenario would be that a single VM needs internet access for downloads, with no need to be reached by other machines.

Bridged Adapter: The vm connected directly with the physical network and acts like it was its own. This is a separate physical machine, with its own IP address separate from your router/DHCP. Only visible to other devices on your LAN. A scenario would be that you want the VM to be reachable from other physical devices like testing a web server from your phone on the same WIFI.

Internal Network: VMs are able to talk to each other on a private virtual network, but this has no connection to the host or the internet at all. A scenario would be building an isolated lab, simulating an air gapped network for security testing, where you don't want any internet exposure.

Host only network: This one is similar to the internal network, but the host machine also can join this private network which still excludes the internet and out lan. A scenario of this is you want the host to manage/access a VM directory without exposing the VM to the rest of the intern or your wider LAN.

4e

1.1: Oldest standard, max of 12MBps, very slow compared to modern alternatives, only suitable for low bandwidth devices like keyboard or mice

2.0: Max speed of 480 mbps, large jump from 1.1, used for most peripherals such as external drives, webcams or printers

3.0: Max theoretical speed of 5 gbps, this is 10x faster roughly than 2.0 and is used for high throughput devices like external SSDs, high res webcams, or fast flash drives.

5a

A core is a physical independent process unit on a chip with its own execution pipeline (registers, cache, etc) and can independently fetch, decode, and execute instructions. More cores will mean more instructions that can actually run in parallel at the same time.

A thread which is known as a logical process is a second set of architectural states with registers and program counters that share the same physical core execution resource. When we have 2 hardware threads on one core it lets the core interleave or overlap instruction from 2 different instruction streams to fill execution slots. They are still using the same underlying cache and pipeline. A core is an independent hardware and a thread is a way of better utilizing a core's existing hardware by giving it 2 views to draw from.

5b

Intel CPU (x86): 86 cores and 172 threads, The Xeon 6 698X costs \$8,469.

AMD CPU (x86): 256 cores and 512 threads. EPYC 9969 cost \$14,904

IBM CPU (Power9): 24 core and 8 threads. S924 system costs \$8,000

ThunderX CPU (ARM): ThunderX3 96 cores and 384 threads cost isn't publicly known.

NVIDIA GPU: 24, 064 cuda cores. RTX PRO 6000 Blackwell costs \$16,000

5c

Clock speed scaling out in the mid 2000s due to the power wall with dynamic power consumption in a chip scales roughly with the square of the voltage and linearly with frequency, so increasing clock speed higher requires more power and generates more heat. Beyond the 5 or 5 GHz current cooling technology makes too much heat per unit chip which becomes nearly impossible to dissipate with current cooling. A Gigahertz plus frequency signal propagation delay across the chip and leakage current.

5d

Moore's law is an observation that the number of transistors on an integrated circuit roughly doubles every 18-24 months while cost per transistor falls, meaning that computing power available per dollar has grown exponential over the years. It will not go on forever because of physical limits, transistors are getting closer to the atomic level, You physically cannot make a transistor smaller than an atom. Expects expect Moore's law scaling to effectively end sometime in 2030, as nodes approach 1 nanometer and below.

6a

Threading enables concurrency which is the ability to make progress on multiple tasks by rapidly switching between them rather than entirely on one block. This matters most a lot of I/O bound work, such that if one thread is waiting on disk then the CPU would otherwise sit and wait idle during that. A second thread can use that idle time to do useful work applications keeping its interface responsive while background threads work.

6b

No, not beyond a certain point that adding more threads can actually hurt performance. Every time the OS switches a thread that is running there's a cost, when too many threads are switching less work is actually being done. Threads compete for the same shared resources so they can slow each other down when you have too many rather than speed things up. If part of a program is inherently sequential, adding threads only speeds up the parallel portion.

6c

For the most part no, theoretically using N threads or cores should speed up a parallel task by at most a factor of N , but since you're splitting the work across more processing units. Super linear speed can happen in a specific narrower circumstance due to cache effects rather than straight up computation. When you split up the dataset across more cores, each core's smaller slice of the problem may not fit completely within that core's local cache.

6d

Locks are needed to help protect the shared and mutable state from race conditions. When a bunch of threads can read and write the same variable or data structure at the same time, the final result can depend on the unpredictable time of which threads executes first leading to potential data loss. An example of this would be if two threads both read a shared counter value and both independently increase their local value of the count and then write back then even though 2 increments happened, the counter only really went up by 1 instead of 2.

6e

Yes because an unbounded number of threads can degrade performance under heavy load due to memory overhead, context switching, or negative returns. The solution is to use a thread pool which is a fixed bounded number of worker threads that will pull tasks away from a queue and instead spawn a new thread per request. This caps resource usage while still getting the benefits of concurrency.